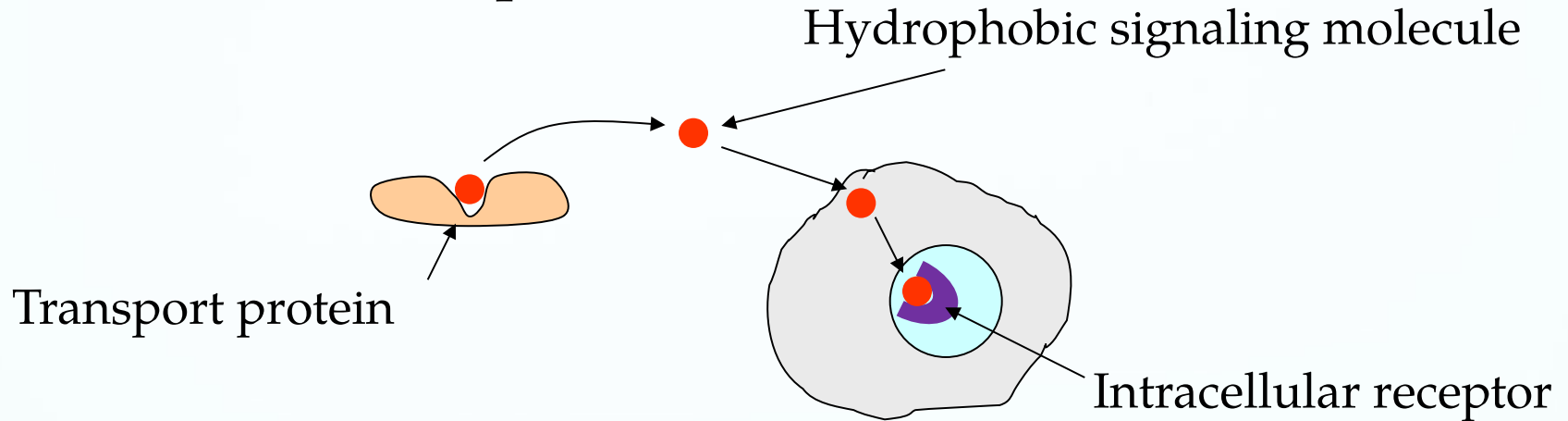


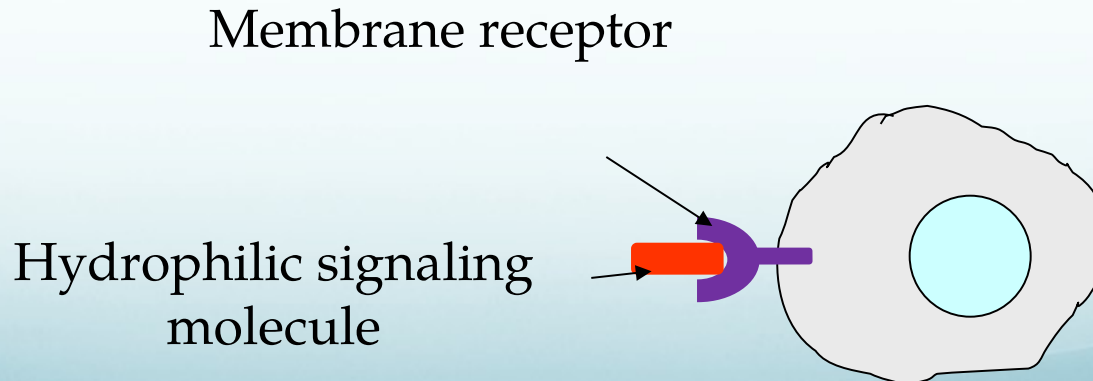
The different types of membrane receptors and how they work

The different types of receptors: 2 big families

- **Intracellular receptors**



- **Membrane receptors**



The different types of membrane receptors are grouped into 3 categories

- ◆ Channel receptors that, after having fixed a chemical messenger, modify the passage of ions inside the cell

Ex: NMDA glutamate receptors

- ◆ Seven-transmembrane receptors or G protein-coupled receptors (GPCR) without any enzyme activity.

- ◆ Single-pass transmembrane receptor displaying an enzymatic activity.

GPCR : Overview

Robert Lefkowitz and Brian Kobilka received the Nobel Prize in Chemistry in 2012 for their work on GPCRs

http://www.youtube.com/watch?v=gPyo7k9E_-w

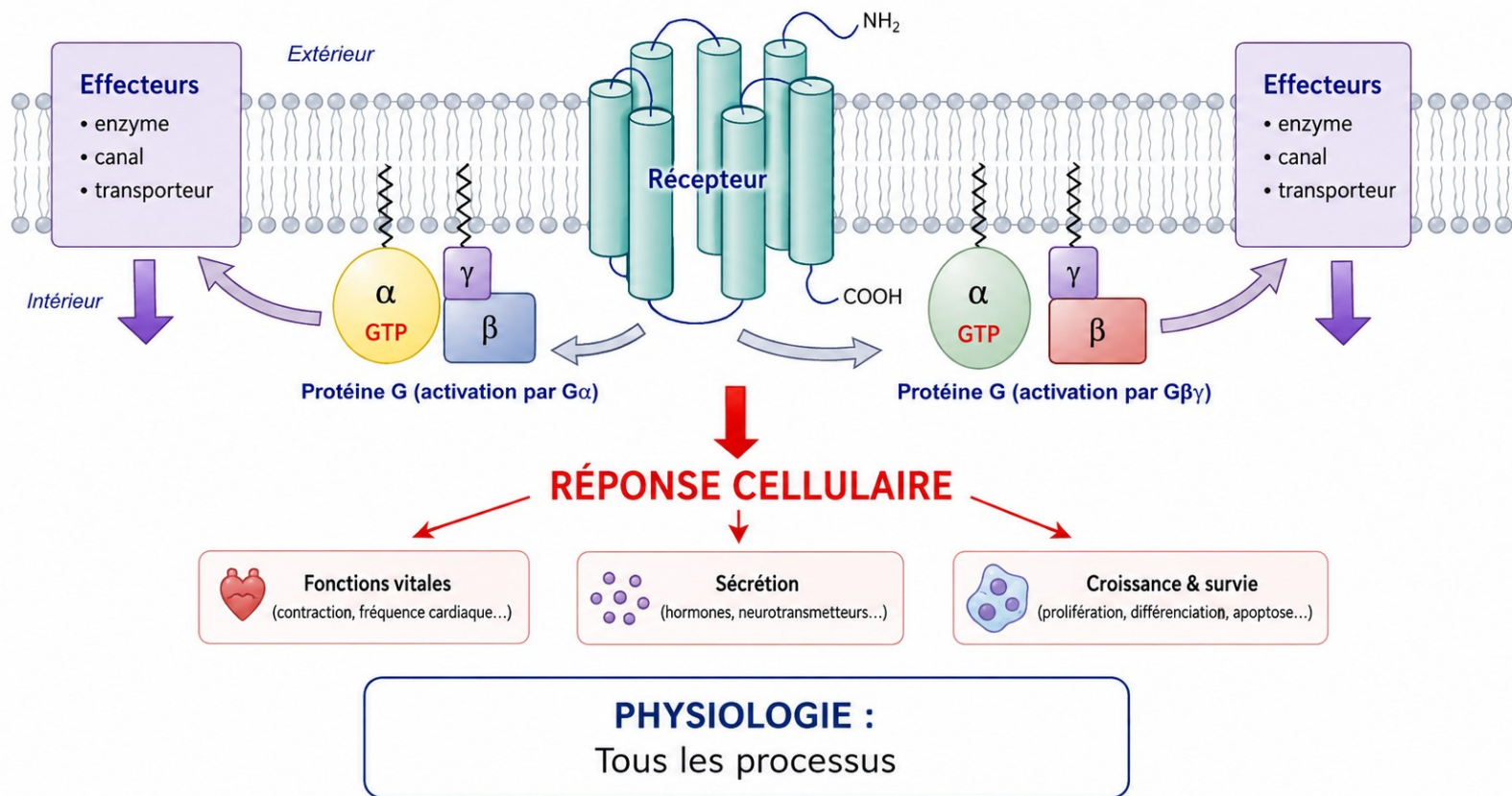
<http://www.youtube.com/watch?v=IP4zsbemW8I>

- More than 1000 GPCR: 500 receptors binds tasting and and odorant molecules
450 endogenous ligand receptors
Some still remain orphans
- The coding sequences of their genes represent more than 1% of the human genome
- 800 genes identified in humans

GPCR : Overview

They are involved in the regulation of all the main functions of the organism

See the list uploaded on moodle



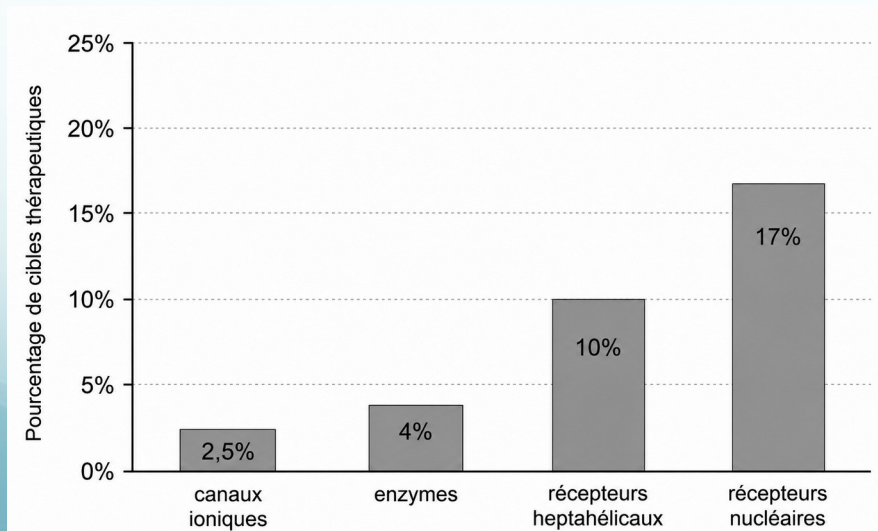
Stimulus	Affected cell type	G Protein	Effector	Effect
Epinephrine Glucagon	Liver cells	G _s	Adenylyl cyclase	Breakdown of glycogen
Epinephrine Glucagon	Fat cells	G _s	Adenylyl cyclase	Breakdown of fat
Luteinizing hormone	Ovarian follicles	G _s	Adenylyl cyclase	Increased synthesis of estrogen & progesterone
Antidiuretic hormone	Kidney cells	G _s	Adenylyl cyclase	Conservation of water by kidney
Acetylcholine	Heart muscle cells	G _i	Potassium channel	Slowed HR & decreased pumping force
Enkephalins endorphine, opioids	Brain neurons	G _i /G _o	Ca ²⁺ & K ⁺ channels, Adenylyl cyclase	Changed electrical activity of neurons
Angiotensin	Smooth muscle cells in blood vessels	G _q	Phospholipase C	Muscle contraction ; elevation of BP
Odorants	Neuroepithelial cells in nose	G _{olf}	Adenylyl cyclase	Detection of odorants
Light	Rod and cone cells in retina	G _t	Cyclic GMP phosphodiesterase	Detection of visual signals
Pheromones	Baker's yeast	GPA1	Unknown	Mating of cells

GPCR :Overview

Because of this, many therapeutic molecules target GPCRs.

- ✦ 3000 substances,
- ✦ 500 targets
- ✦ 26.8% of drugs
target GPCRs

Among the different types of receptors, 10% of GPCRs are exploited targets ...



GPCR : Overview

Drugs that target a GPCR treat many pathologies

Asthma

Serevent/ Ventolin



Diabetes insipidus

Desmopressin



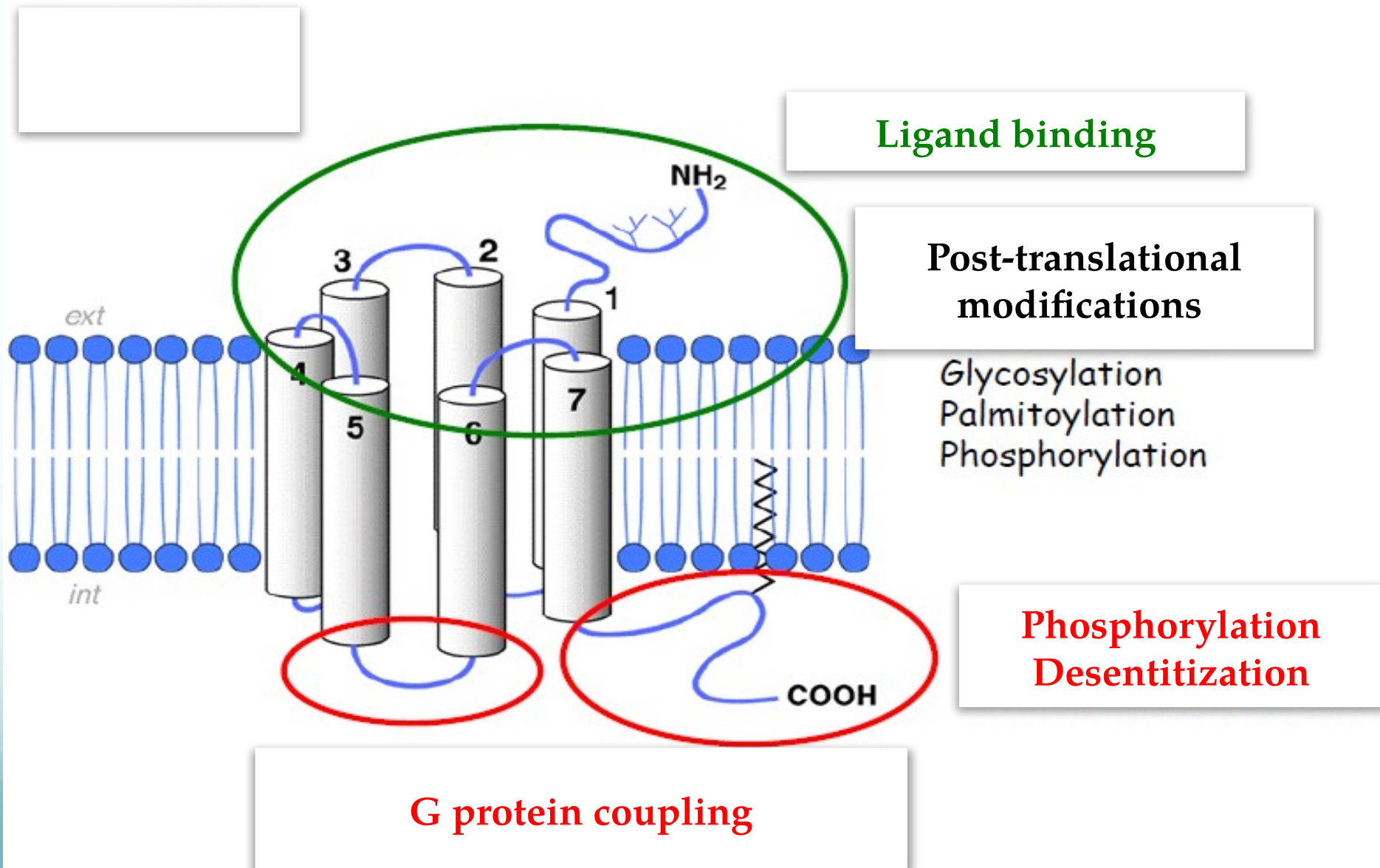
**A
F
D
I**

Hypertension

Acebutolol



GPCR : general overview



GPCR : Overview

- Up to 2008, only two GPCRs were crystallizing :

- Rhodopsin (2000)

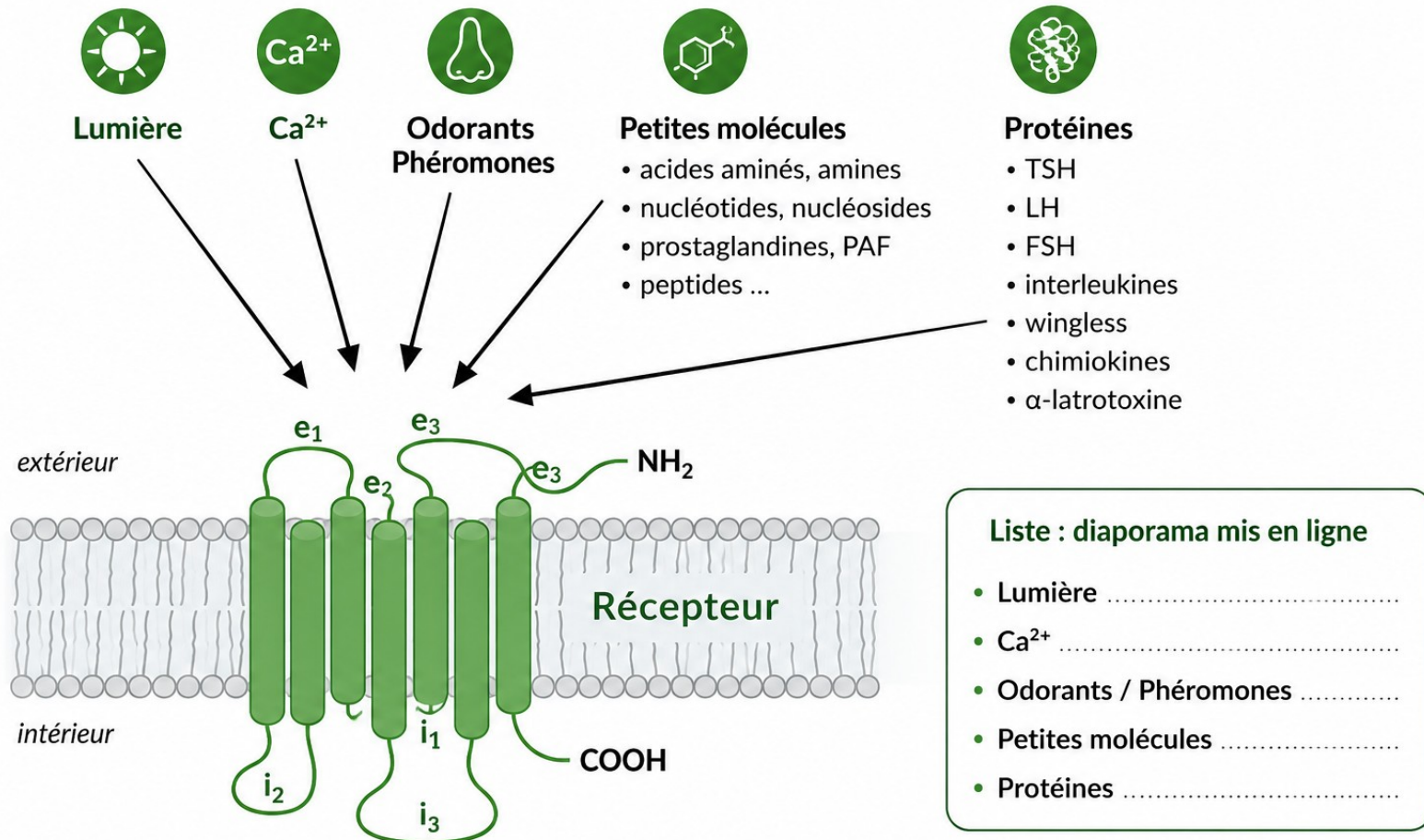
- The β 2-adrenergic receptor (2008)

Since then, several other GPCR have been crystallizing (Human A2AR, NTSR1, A2AR, CXCR1).

GPCR : Overview

GPCR bind all kind of different messenger molecules

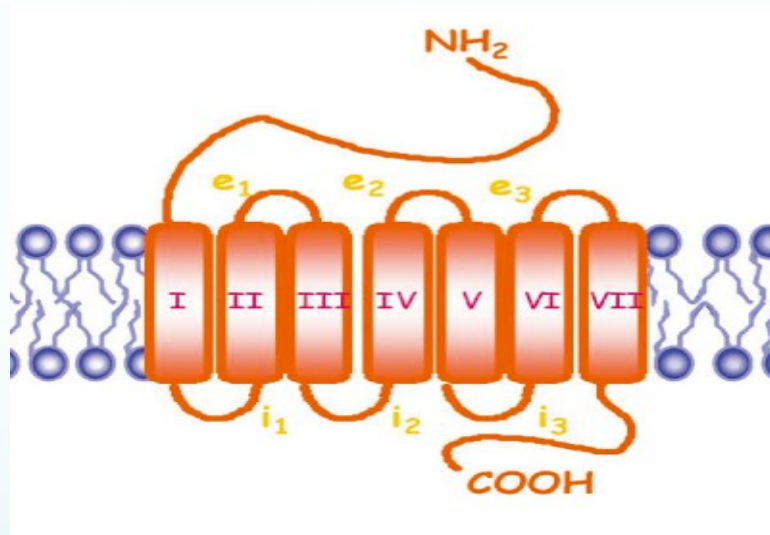
A



GPCR : Overview

The initial classification (1994): based only on similarities sequences ...

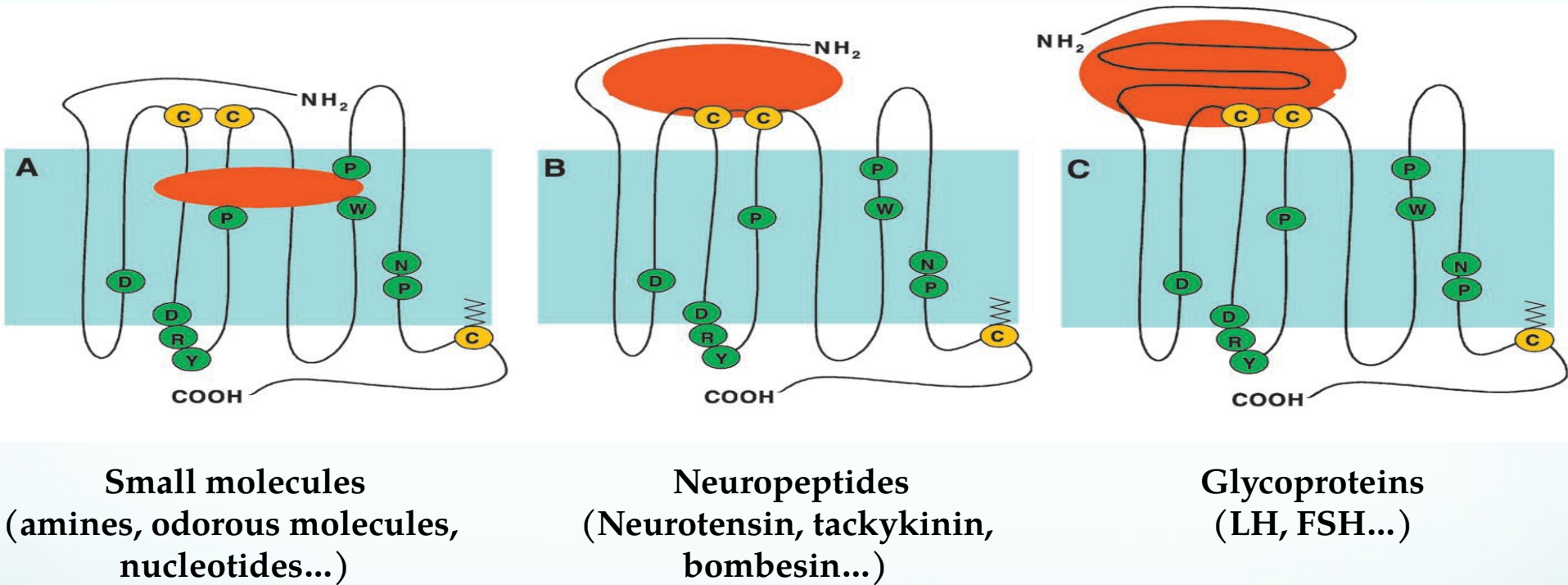
Currently, the most used classification is based on sequence homologies, ligand binding modality and the size of the N terminal domain.



most mammalian GPCRs can be grouped in 3 families

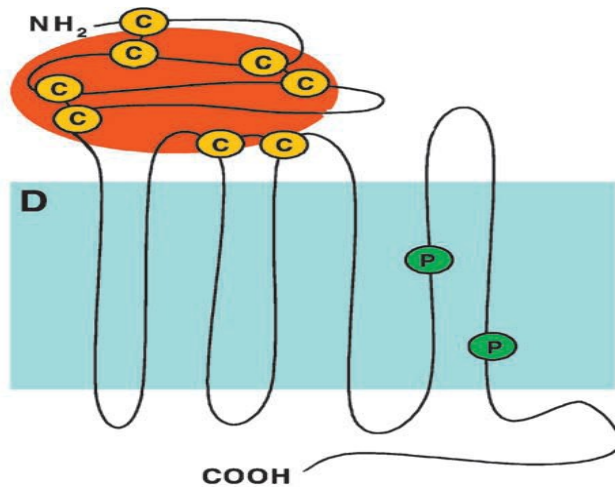
GPCR : general information

Family 1



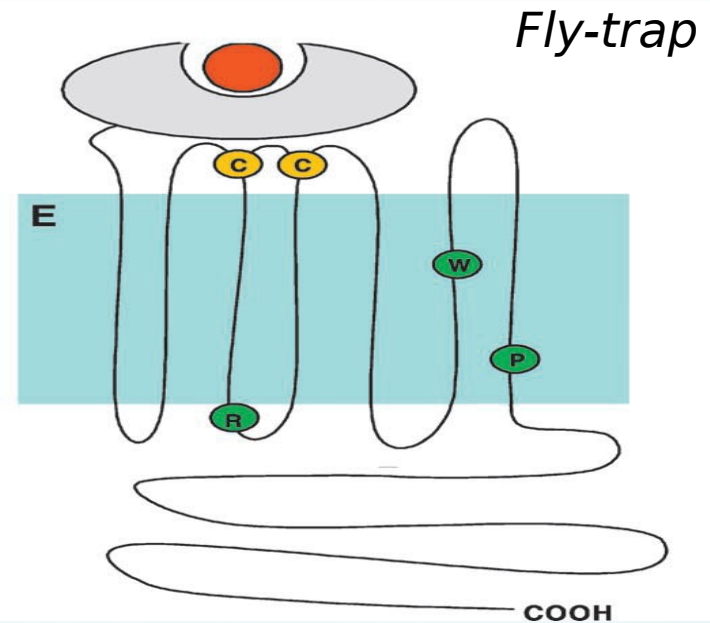
GPCR : general information

Family 2



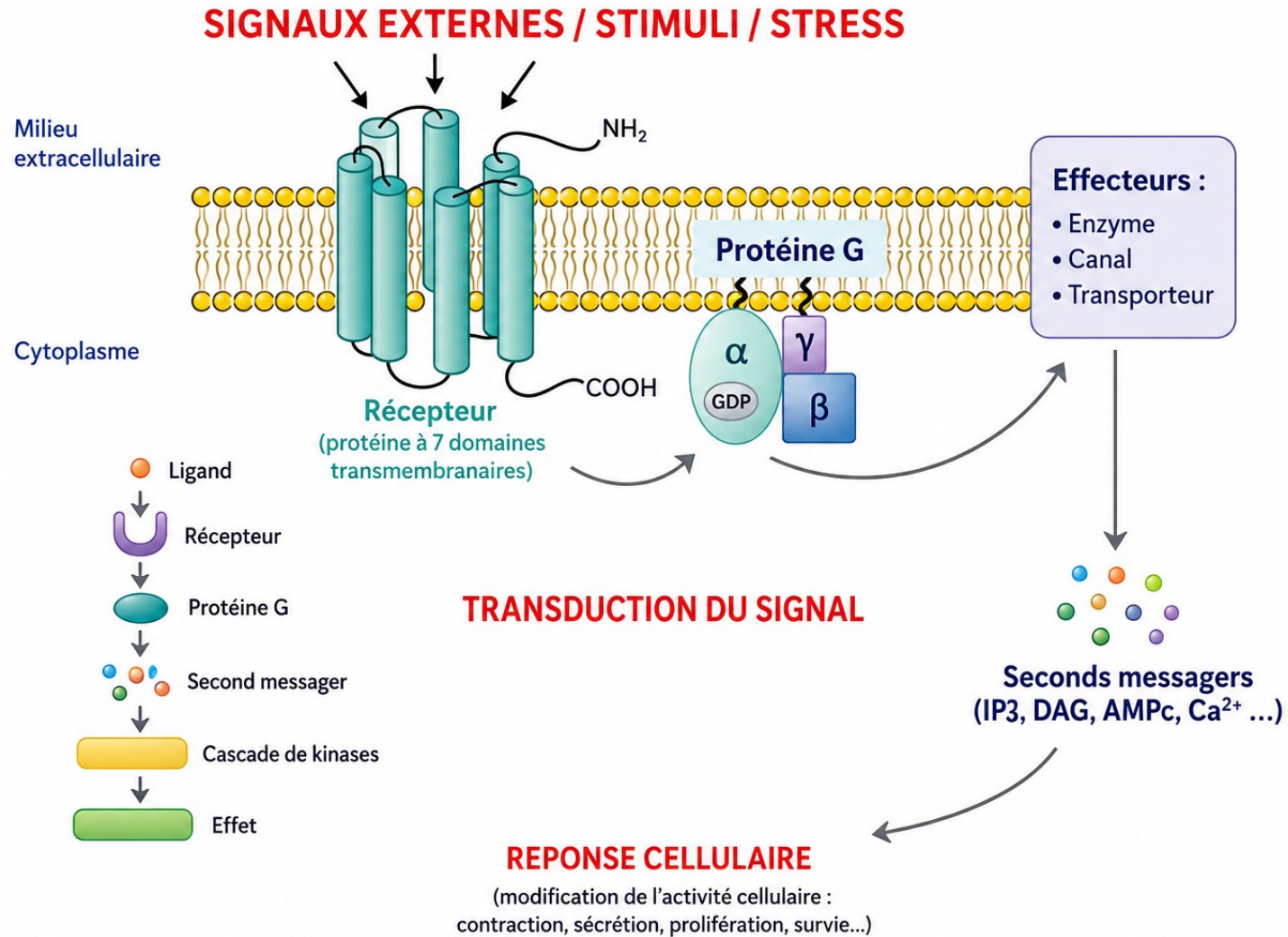
Peptidic hormones
(secretin,
glucagon, GnRH, VIP, PACAP...)

Family 3



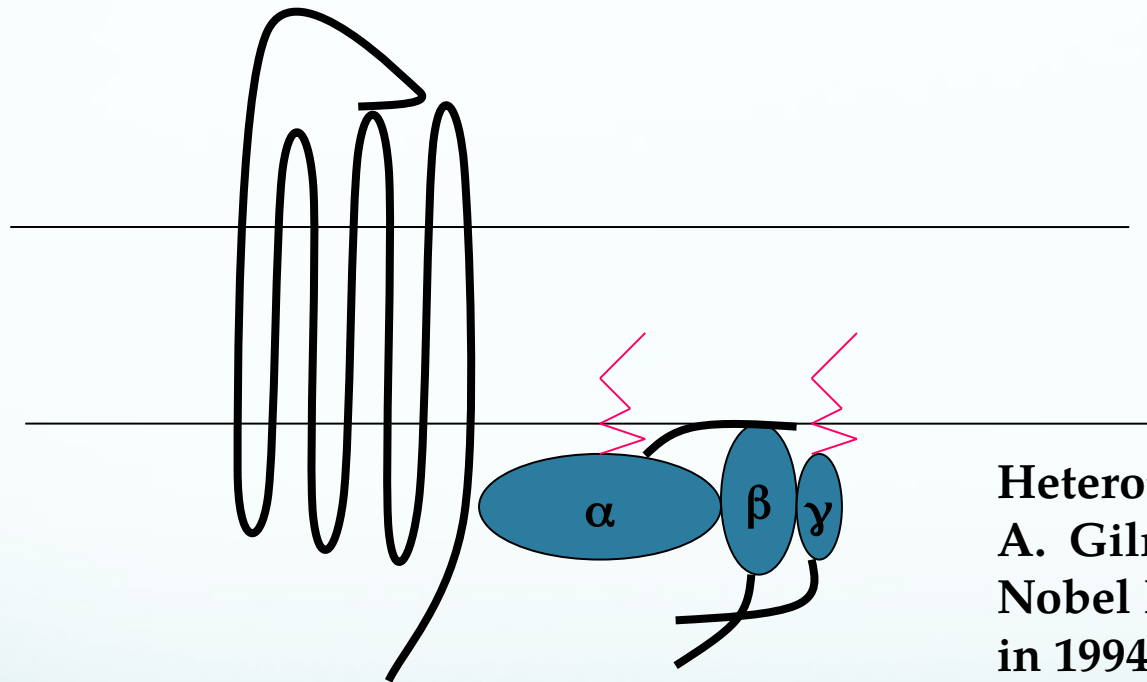
Glutamate, GABA, pheromones

GPCR: signal transduction



GPCR: signal transduction

- *Via* heterotrimeric G proteins

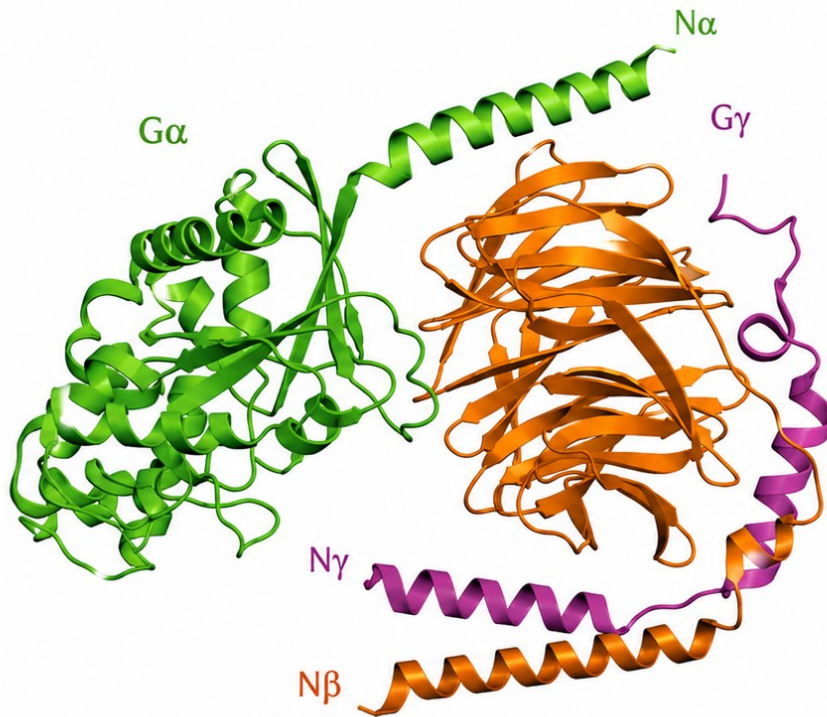


Heterotrimeric G proteins :
A. Gilman et M. Rodbell B.
Nobel Price of de medicine
in 1994

- Independently from heterotrimeric G proteins

GPCR: signal transduction

- *Via* heterotrimeric G proteins



	Masse Mol. (kDa)	Propriétés
α (20)	39–52	• lie GDP ou GTP • activité GTPasique • parfois ADP-ribosylables
β (5)	36	} • toujours associées
γ (13)	7–8	

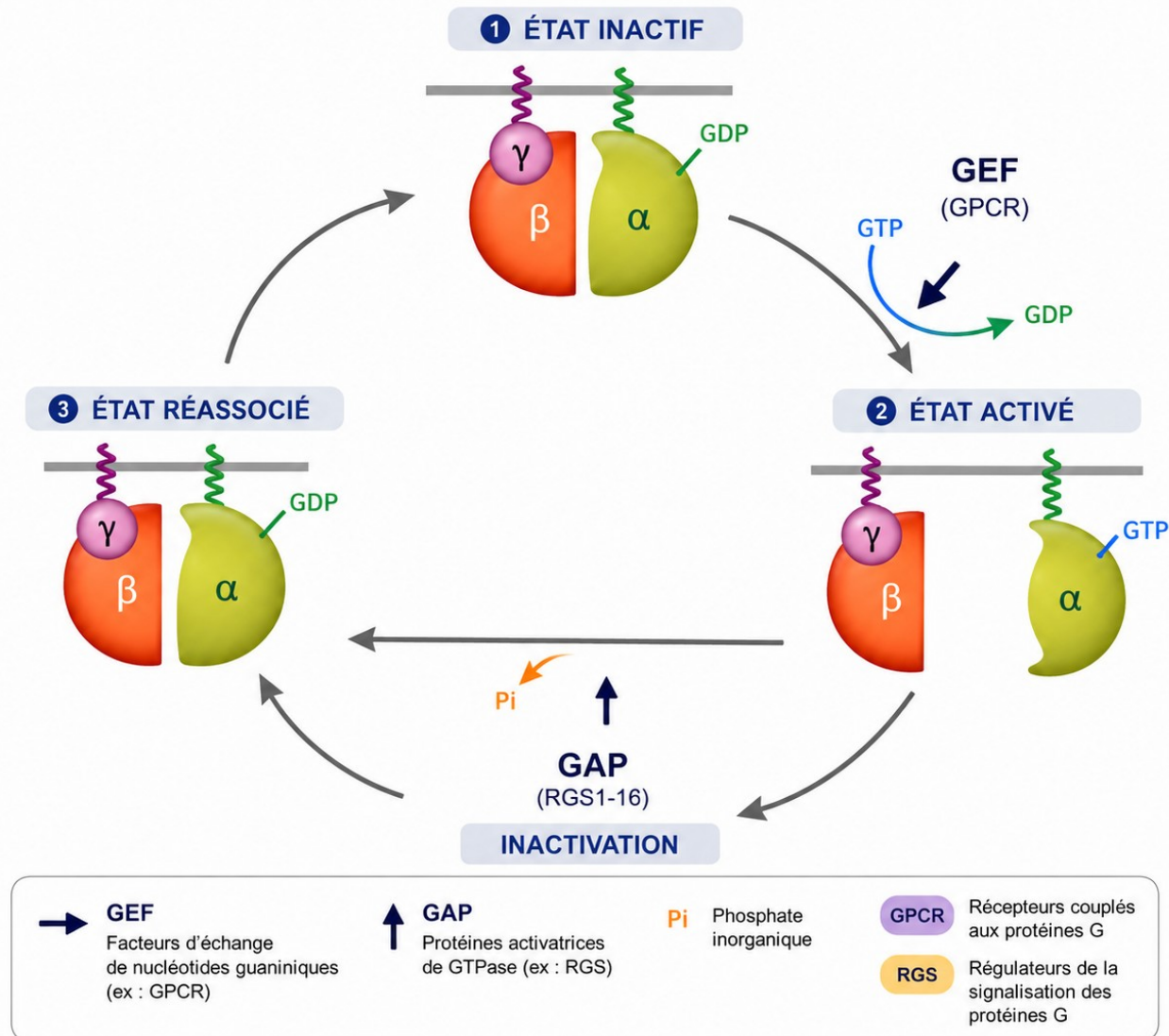
Figure 2. Structure tridimensionnelle du complexe hétérotrimérique $G\alpha_{GDP}-G\beta-\gamma$ (PDB code 1GOT).

La sous-unité α est représentée en vert et les sous-unités β et γ en orange et magenta respectivement.

The interaction between β and γ is stabilized by a superhelical type of interaction (Garritsen et al., 1993).

GPCR: signal transduction

- *Via* heterotrimeric G proteins



Cycle d'activation d'une protéine G hétérotrimérique

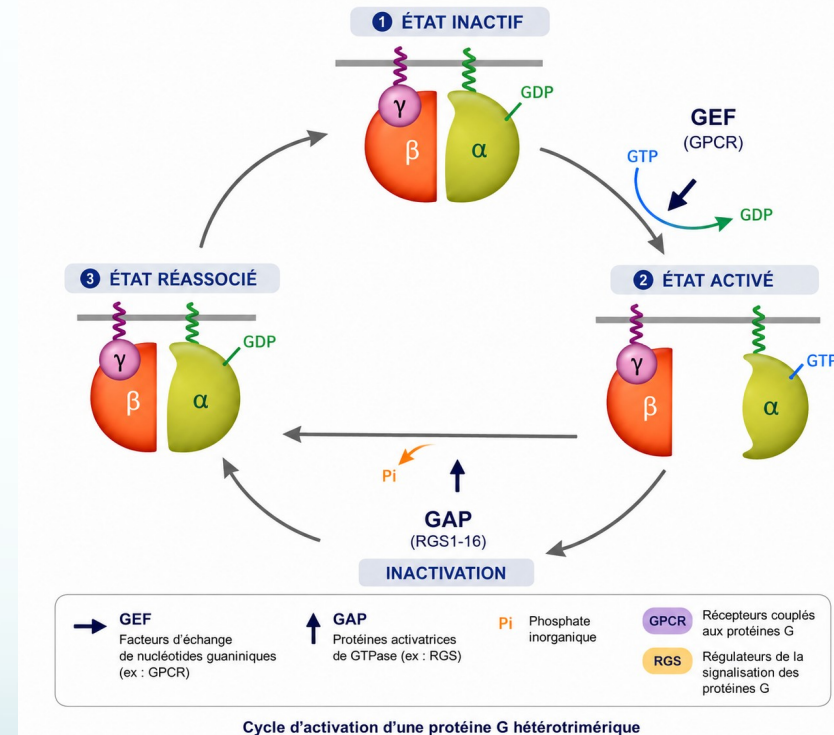
Just a small aside...

GPCRs are not the only ones in the life of G proteins

...

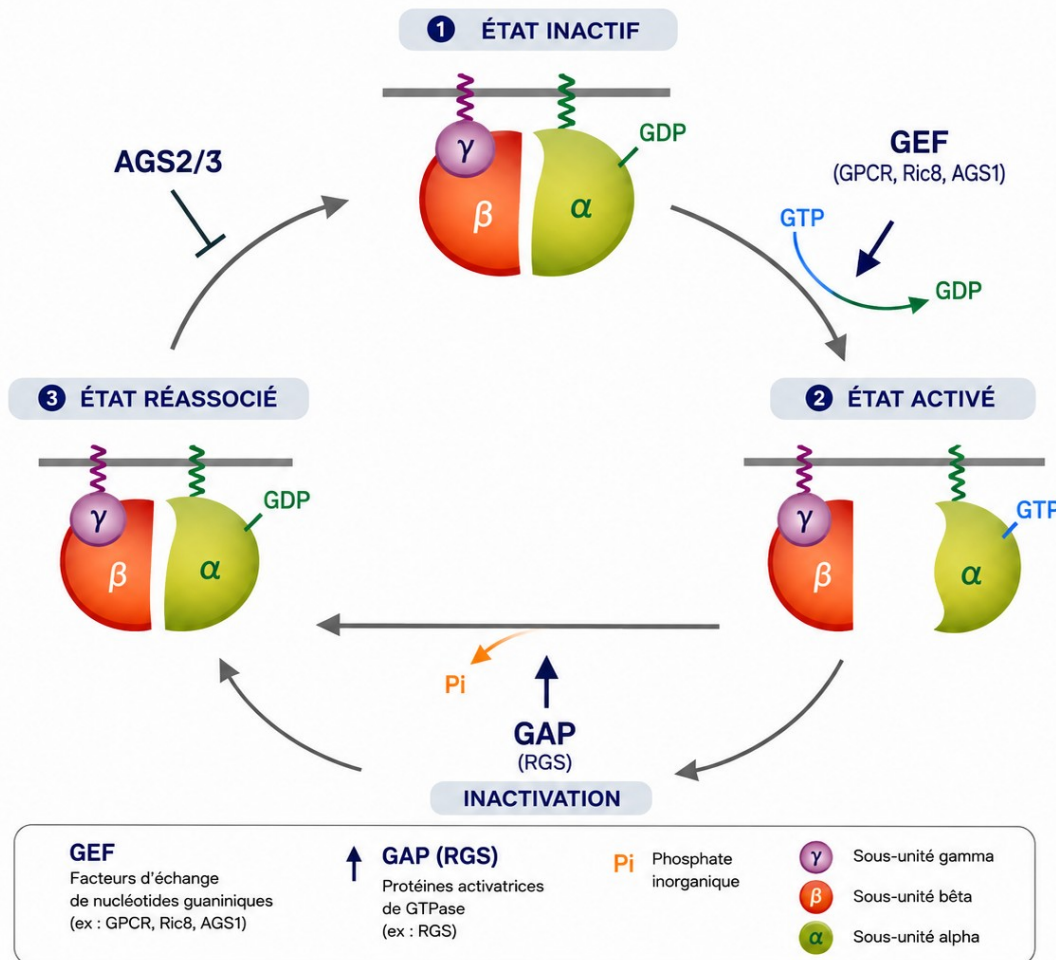
Heterotrimeric G proteins activation

- by other membrane receptors (tyrosine kinase receptors such as the insulin receptor)
- by cytosolic factors such as receptor-independent Activators of G protein Signaling.



GPCR: signal transduction

- *Via* heterotrimeric G proteins



(R)egulator of (G) protein (S)ignaling
€ à la famille des **GTPase-Activating Protein**

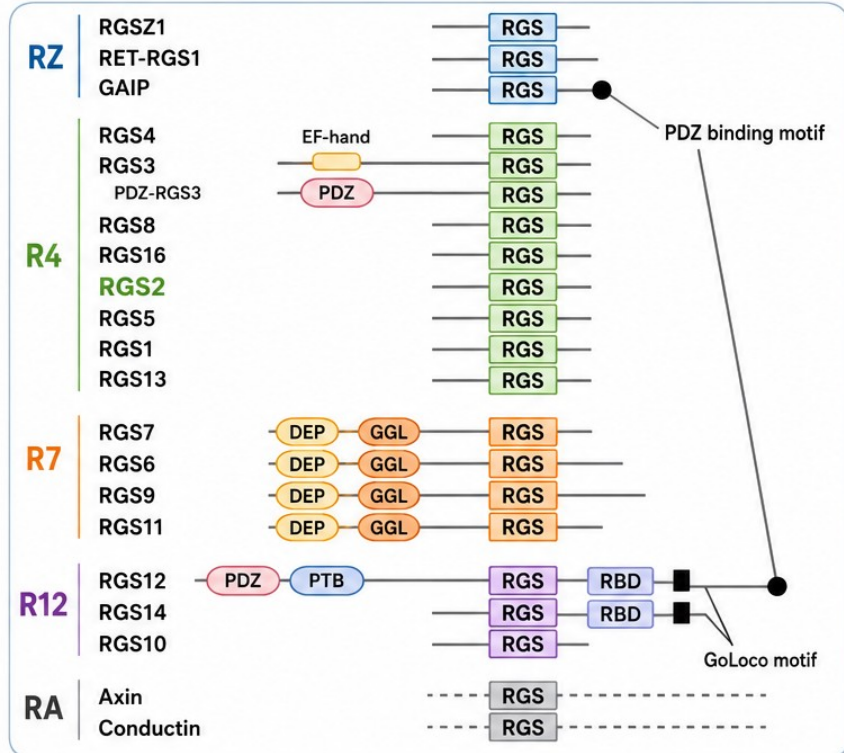
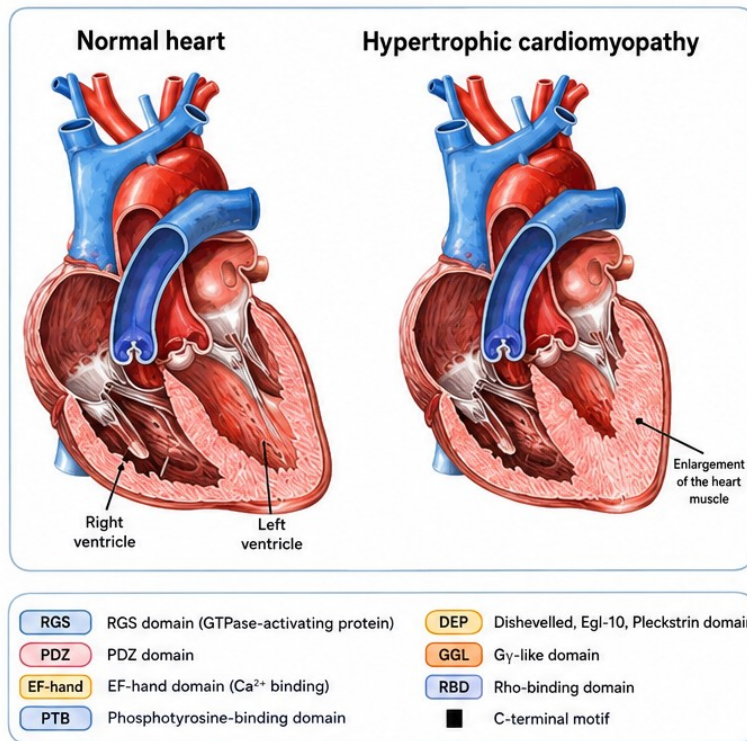
Selective Loss of Fine Tuning of $G_{\alpha 11}$ Signaling by RGS2 Protein Exacerbates Cardiomyocyte Hypertrophy*

Received for publication, July 20, 2005, and in revised form, December 14, 2005

Published, JBC Papers in Press, December 26, 2005, DOI 10.1074/jbc.M500878200

Wei Zhang, Thomas Anger, Jialin Su, Jianming Hao, Xiaomei Xu, Ming Zhu, Agnieszka Gach, Lei Cui, Ronglih Liao, and Ulrike Mende

From the Cardiovascular Division, Brigham and Women's Hospital and Harvard Medical School, Boston, Massachusetts 02115



GPCR: signal transduction

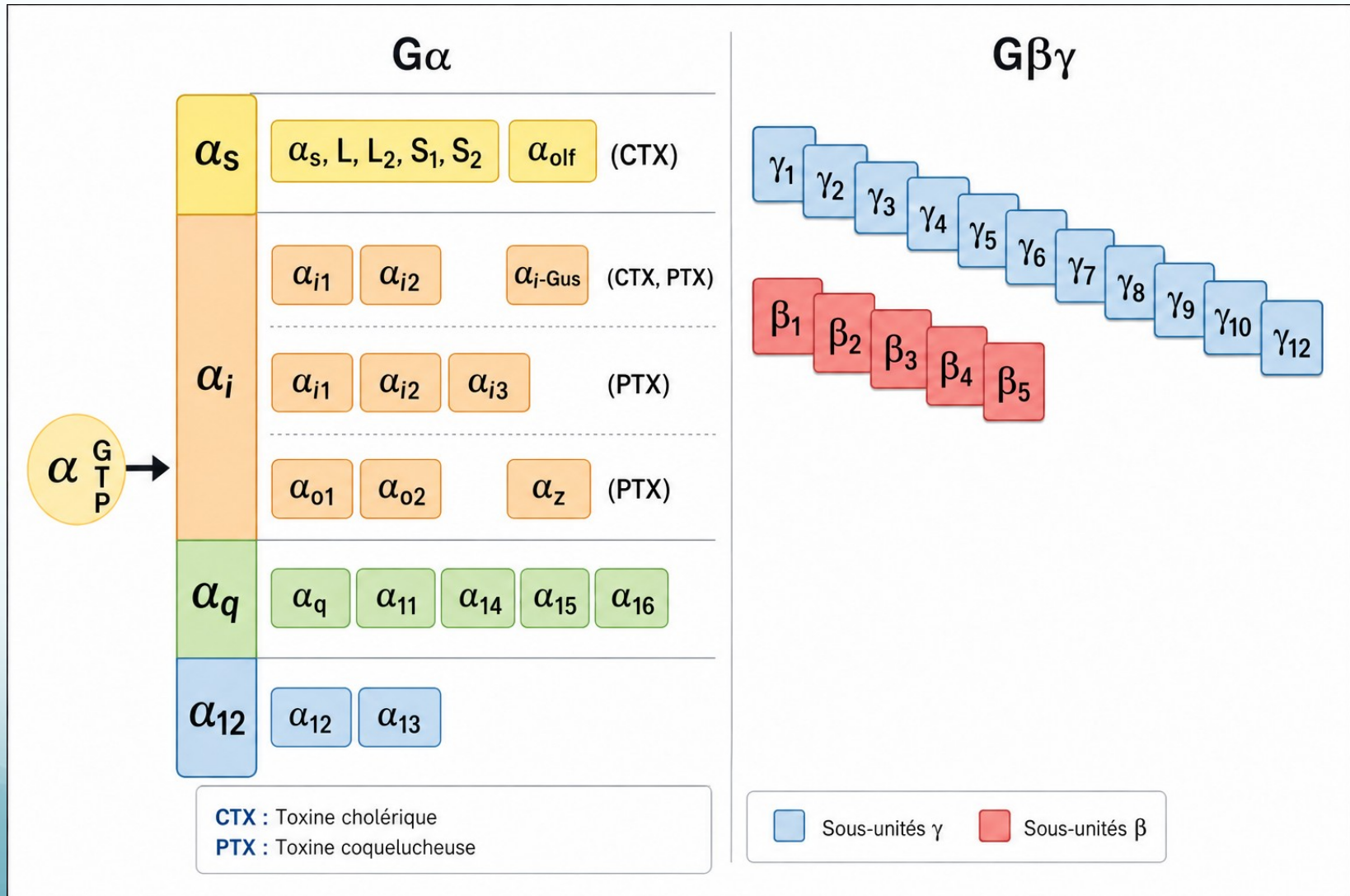
- *Via* heterotrimeric G proteins

17 genes encoding G α , 5 G β and 12 G γ
some of which are subject to alternative splicing

According to a simple combinatorial calculation,
1020 different heterotrimers could exist if each paired randomly with the other 2.

GPCR: signal transduction

- *Via* heterotrimeric G proteins
- Grouped in 4 classes



GPCR: signal transduction

The various classes of subunits α :

◆ Gs ("s" for stimulator of adenylyl cyclase):

- 4 types of α subunits, resulting from alternative splicing
- Golf (olfactory) subunit, mainly for olfactory receptors.

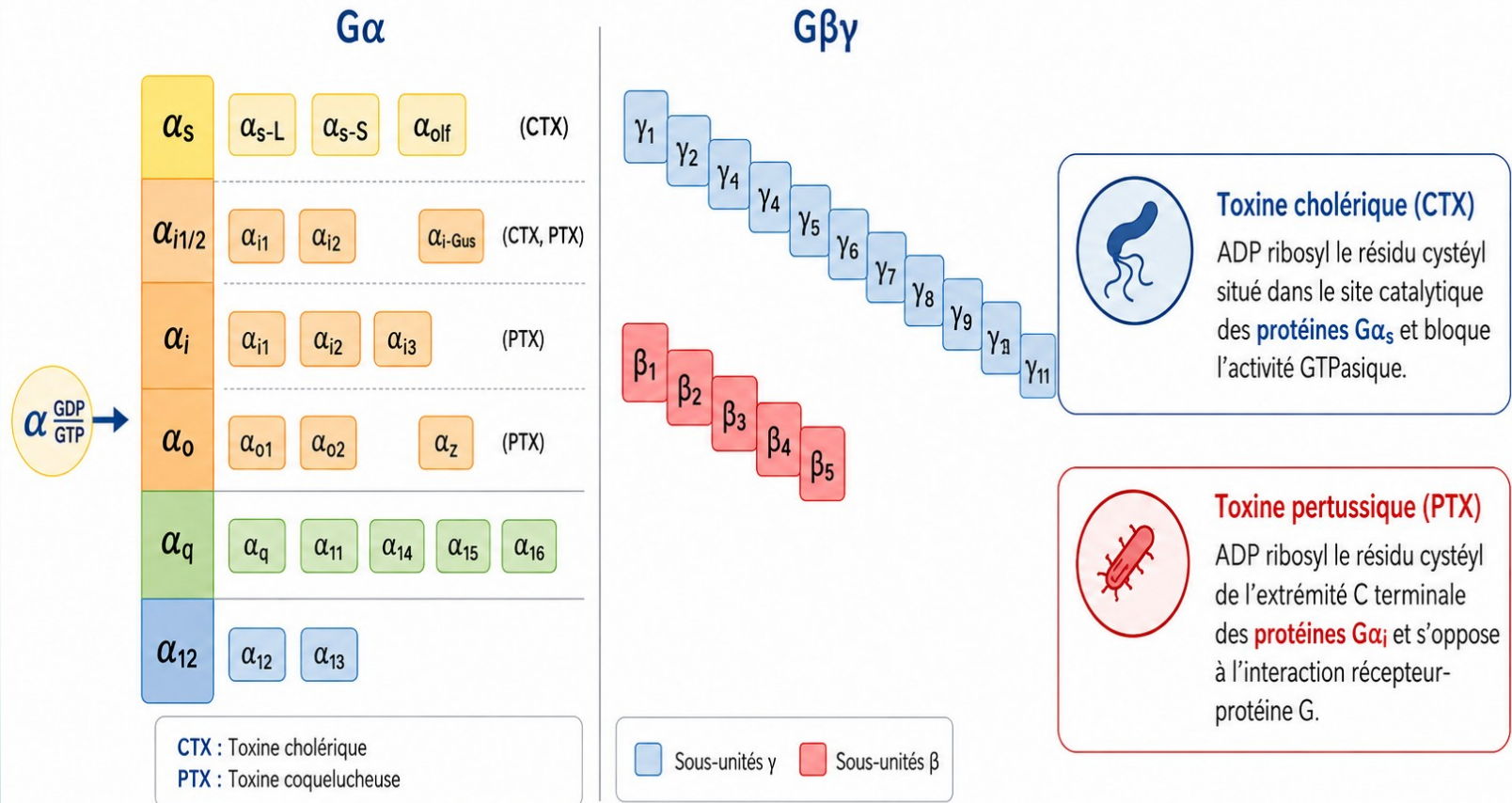
◆ Gi/o ("i" for adenylyl cyclase inhibitor and "o" for others)

- 3 Gi ($\alpha i1$, $\alpha i2$, $\alpha i3$)
- 2 types of G transducin ($\alpha t1$, $\alpha t2$ - rod and cone cells of the retina) from alternative splicing
- 1 G gustducine (αgus)
- 2 Go ($\alpha o1$, $\alpha o2$)
- 1 Gz (αz)

◆ Gq/11 (phospholipase C stimulators): $G\alpha q$, $G\alpha 11$, $G\alpha 14$, $G\alpha 15$ and $G\alpha 16$

◆ G12 (regulation of the cytoskeleton and certain processes linked to movement): the $G\alpha 12$ and $G\alpha 13$ subunits

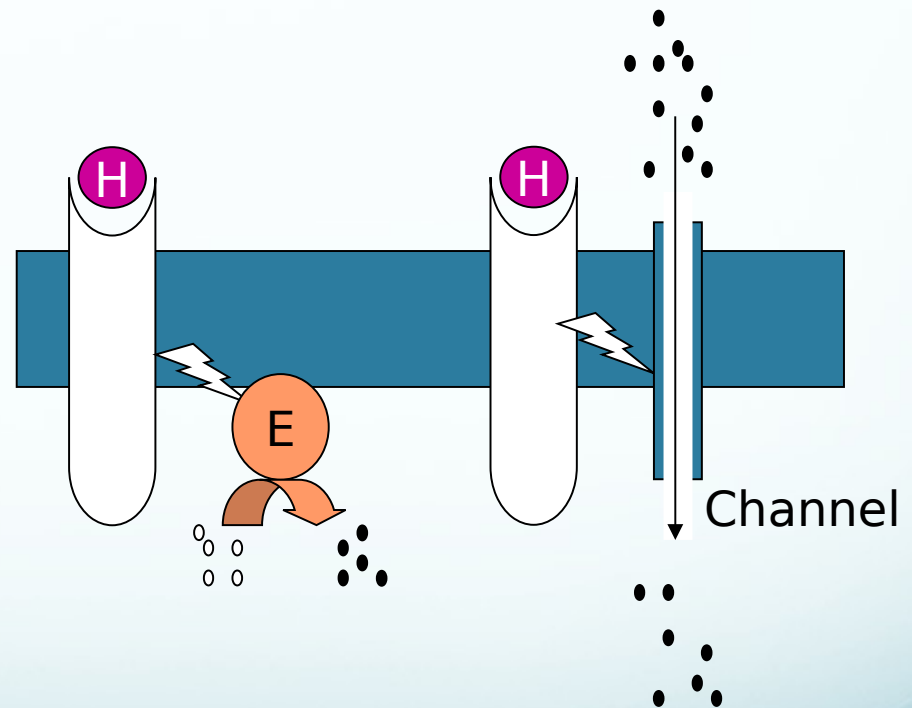
Les RCPG: transduction du signal



GPCR: signal transduction

Intracellular effectors regulated by the different $G\alpha$ and $G\beta\gamma$ subunits

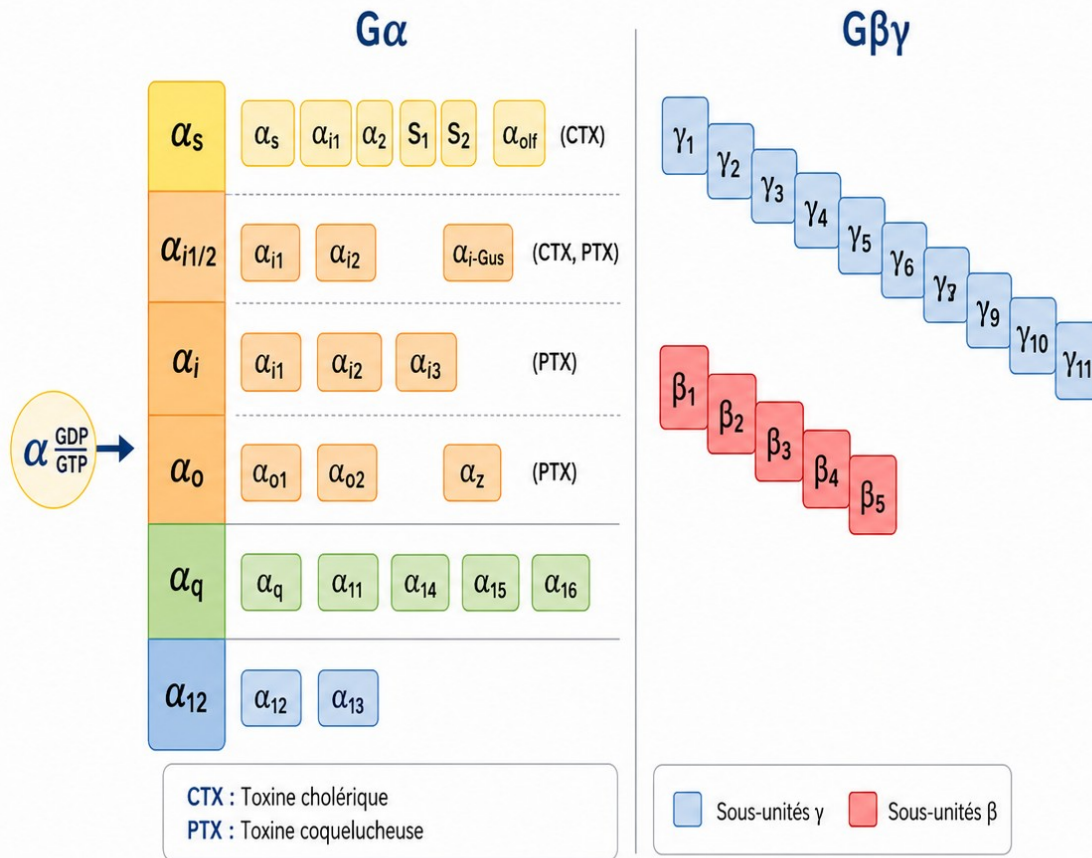
G-protein-family	α -subunit	Gene	Signal transduction
G_i-family			
$G_{i/o}$	α_i, α_o	GNAO1 , GNAI1 , GNAI2 , GNAI3	Inhibition of adenylate cyclase , opens K^+ -channels (via β/γ subunits), closes Ca^{2+} -channels
G_t	α_t (Transducin)	GNAT1 , GNAT2	Activation of phosphodiesterase 6
G_{gust}	α_{gust} (Gustducin)	GNAT3	Activation of phosphodiesterase 6
G_z	α_z	GNAZ	Inhibition of adenylate cyclase
G_s-family			
G_s	α_s	GNAS	Activation of adenylate cyclase
G_{olf}	α_{olf}	GNAL	Activation of adenylate cyclase
G_q-family			
G_q	$\alpha_q, \alpha_{11}, \alpha_{14}, \alpha_{15}, \alpha_{16}$	GNAQ , GNA11 , GNA14 , GNA15	Activation of phospholipase C
$G_{12/13}$-family			
$G_{12/13}$	α_{12}, α_{13}	GNA12 , GNA13	Activation of the Rho family of GTPases



$G\beta\gamma$

KIR3.1–3.4 (canaux K^+ : GIRK)
 GRKs
 Adénylate cyclases (ACII, ACIV)
 Phospholipases (PLC $\beta_1, \beta_2, \beta_3, PL A_2$)
 PI3K γ

GPCR: signal transduction



Toxine cholérique (CTX)

ADP ribosyl le résidu cystéyl situé dans le site catalytique des **protéines α_s** et bloque l'activité GTPasique.



Toxine pertussique (PTX)

ADP ribosyl le résidu cystéyl de l'extrémité C terminale des **protéines α_i** et s'oppose à l'interaction récepteur-protéine G.

GPCR: signal transduction

TD: Examples of signal transduction by addressing cAMP, cGMP signaling and calcium

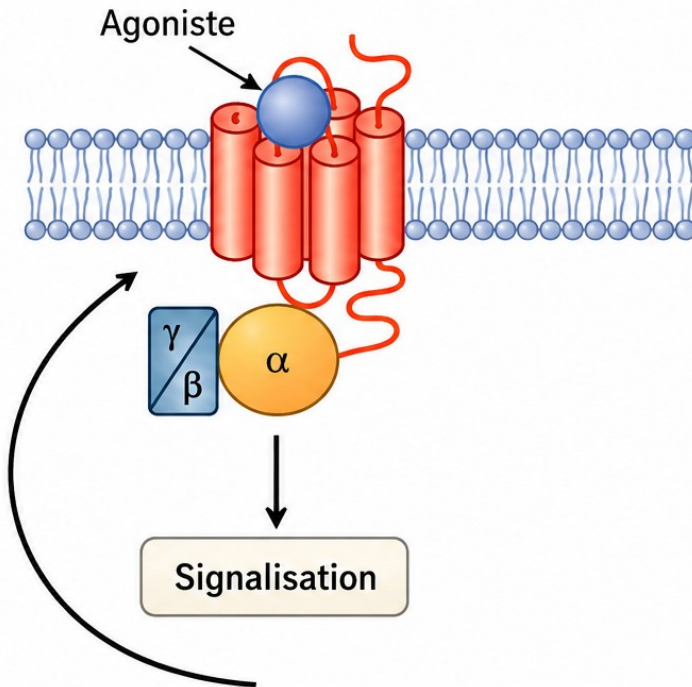
GPCR: stop of signal transduction

The cellular responses generated by GPCRs are rapidly attenuated. This stop of the signal results from mechanisms that operate at the different level of the signalitic cascade.

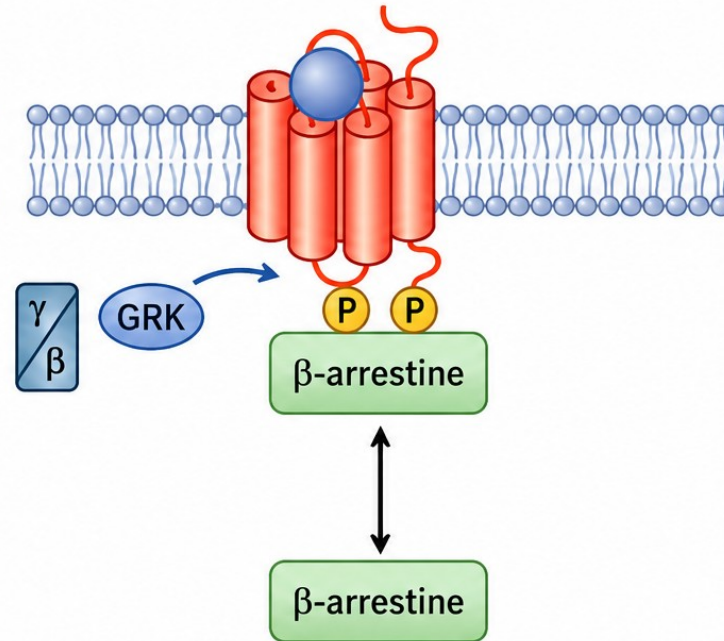
- Suppression of the agonist from the extracellular environment
 - recapture (at the synaptic level)
 - extracellular degradation of the agonist (peptidases)
- Desensitization of GPCRs
 - phosphorylation by « G-protein receptor kinases » (GRK)

GPCR: G protein-dependent stop of signal transduction

(a) Couplage récepteurs – protéines G



(b) Désensibilisation – découplage des protéines G



G $\beta\gamma$

KIR3.1–3.4 (canaux K⁺ ; GIRK)

GRKs

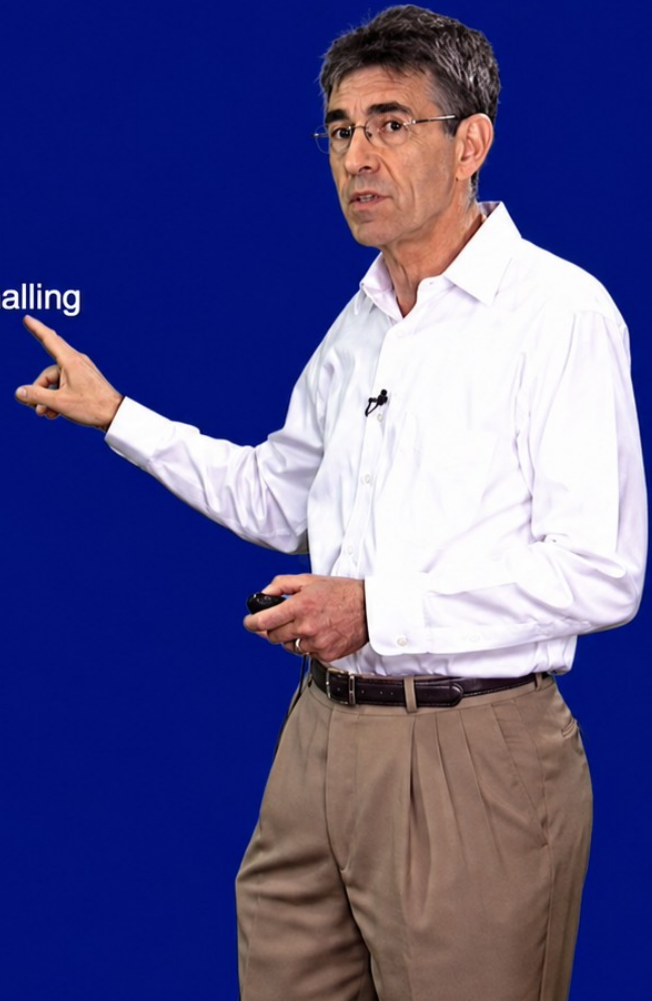
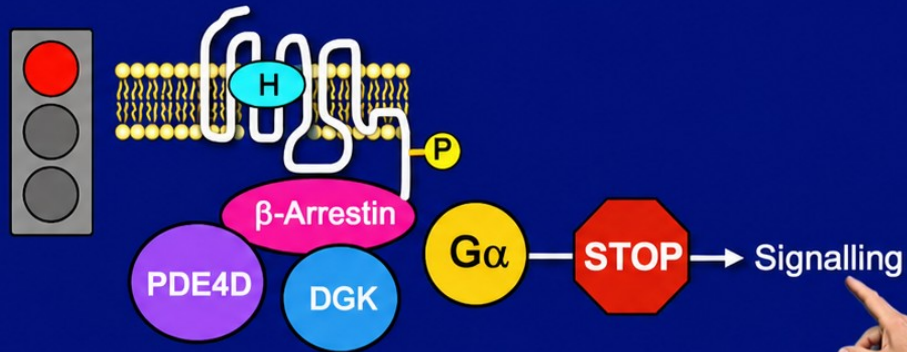
Adénylate cyclases (ACII, ACIV)

Phospholipases (PLC β_1 , β_2 , β_3 , PL A₂)

PI3K γ

β -arrestins: Receptor Traffic Cops

DESENSITIZATION



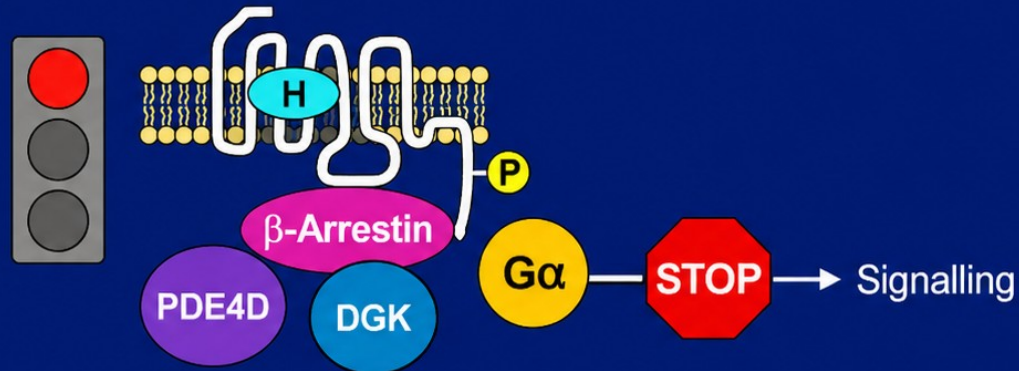
GPCR: G protein-dependent stop of signal transduction

Functional significance of desensitization

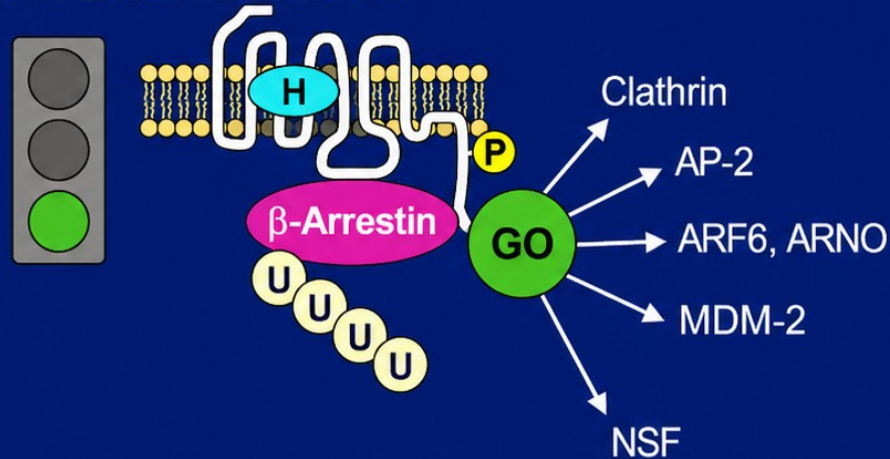
- limit the signal in time: fine regulation of the signal;
- Avoid stimulation overload and its toxic effects;
- In contrast, the effect of agonists is limited.

β -arrestins: Receptor Traffic Cops

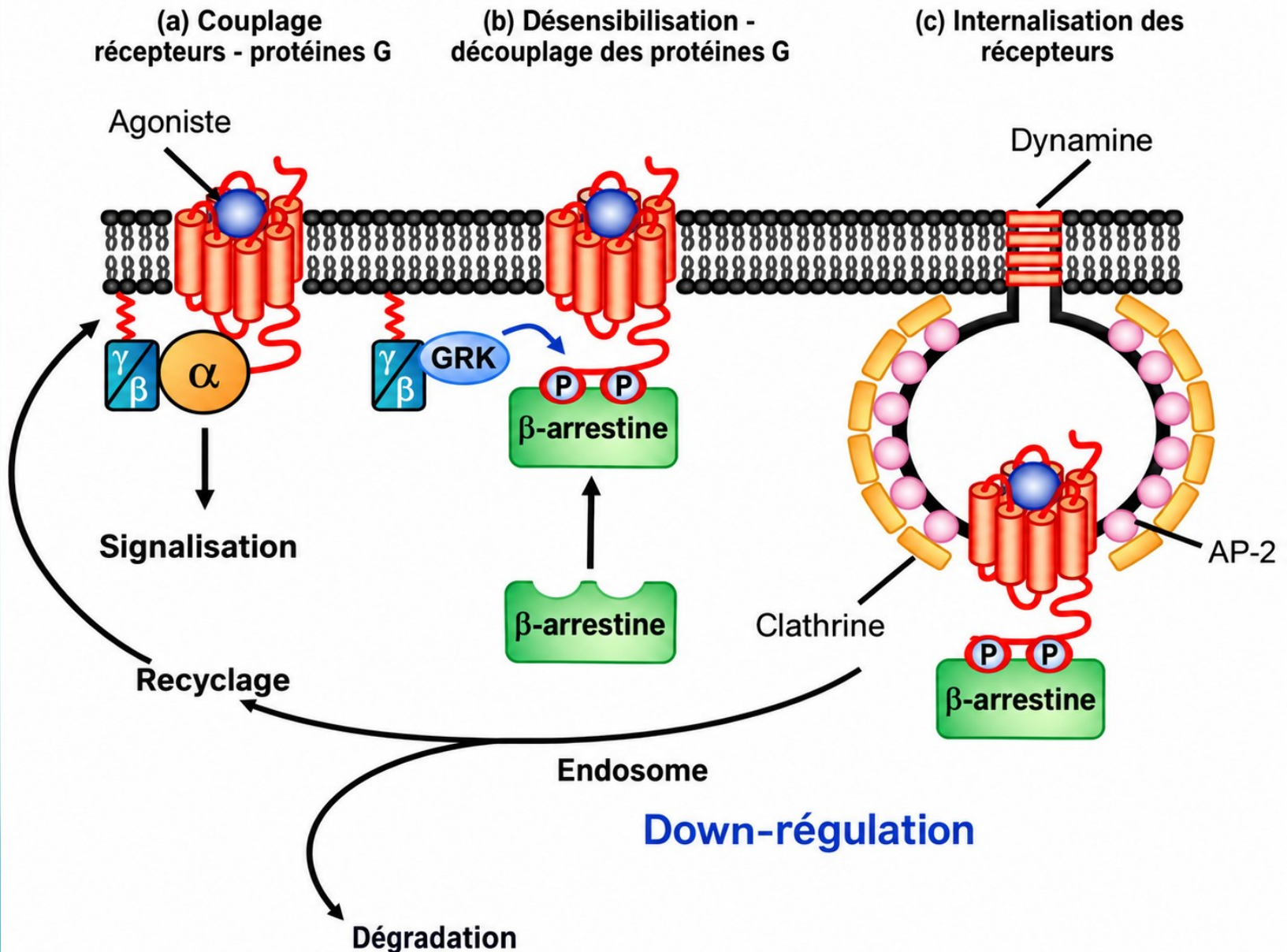
DESENSITIZATION



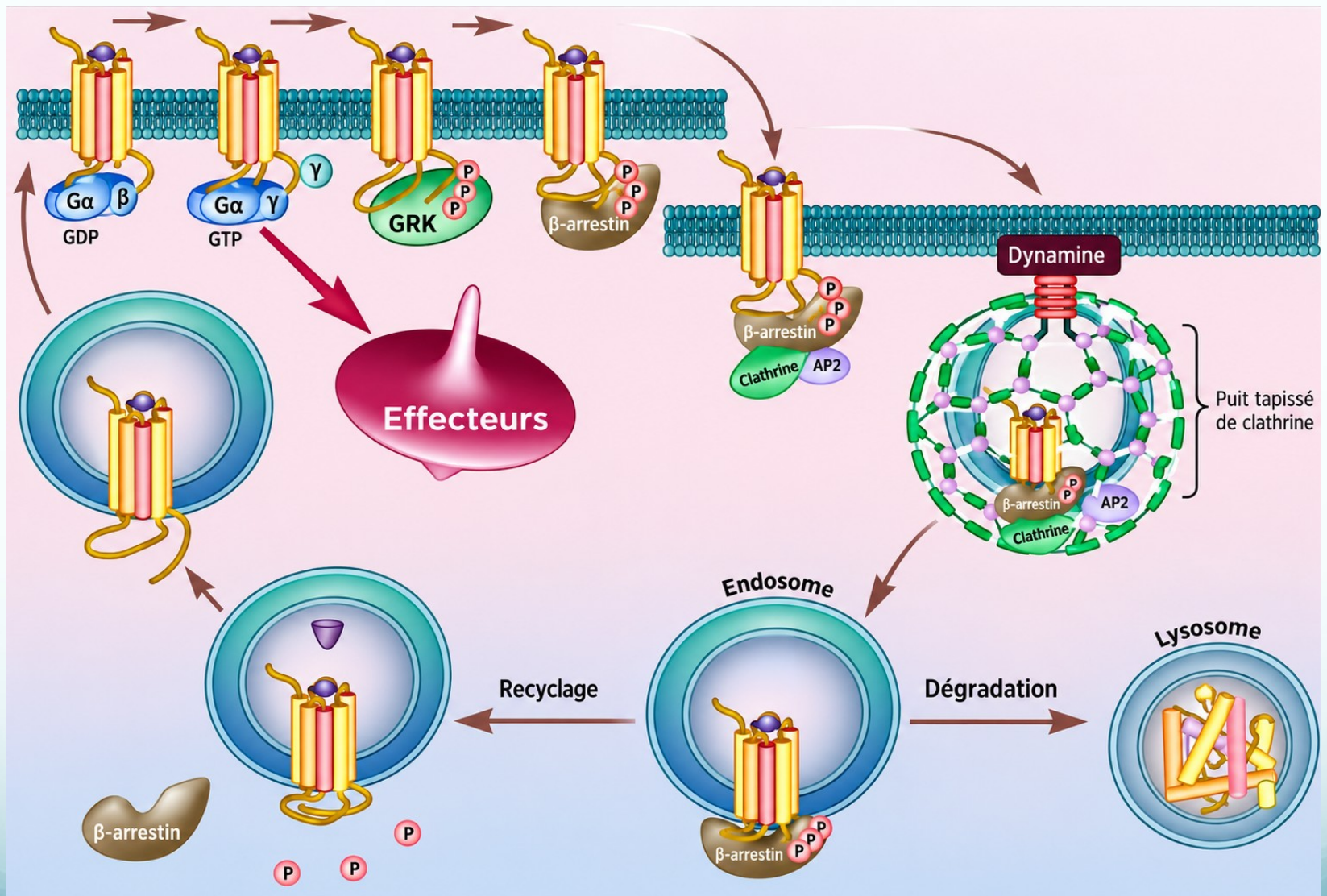
INTERNALIZATION



GPCR: G protein-dependent stop of signal transduction

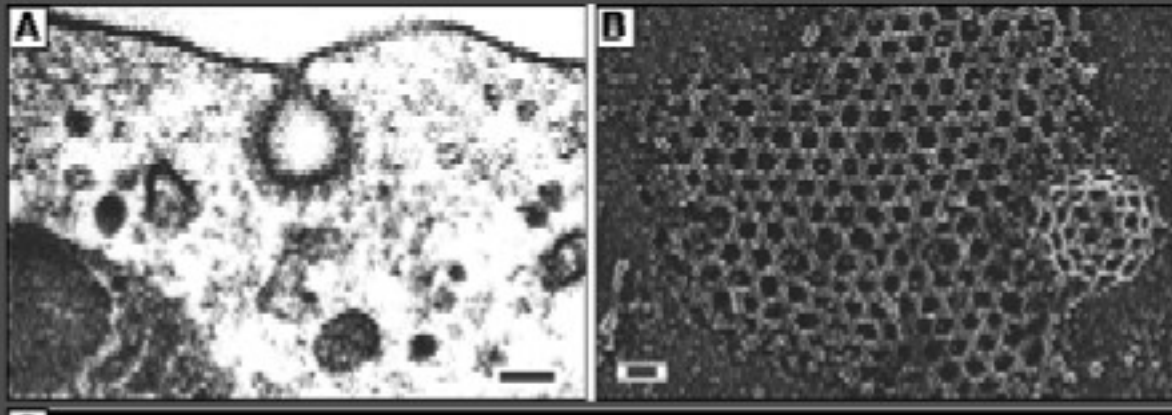


GPCR: G protein-dependent stop of signal transduction



GPCR: G protein-dependent stop of signal transduction

Clathrin-coated vesicles



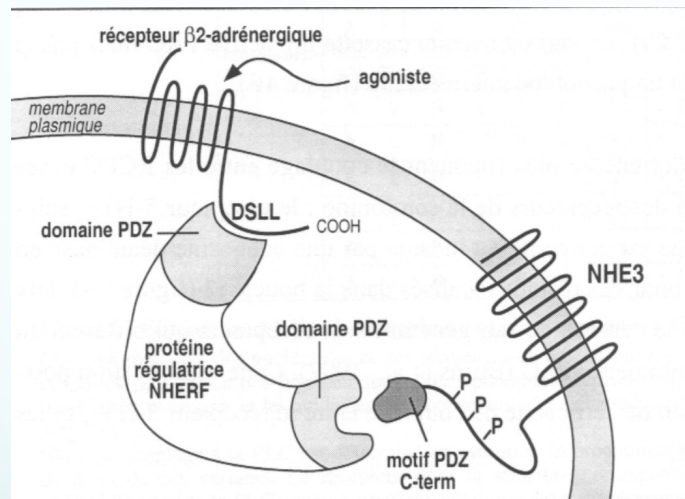
Electronic microscopy

Since the discovery of the GRK and arrestins...

Pathologie	Modification observée	Récepteur impliqué	Piste thérapeutique
Maladie d'Oguchi [6, 7]	Mutation de la GRK1 et de l'arrestine 1	Rhodopsine	Rétablir l'expression de la GRK1 et de l'arrestine 1
Défaillance cardiaque chronique [33, 34]	Augmentation de l'expression de la GRK2	Récepteurs β_1 -adrénergiques	Inhibition de l'activité GRK2 cardiaque
Hypertension [9, 17]	Augmentation de l'expression de la GRK2	Récepteurs β -adrénergiques	Inhibition de l'activité GRK2 des muscles lisses vasculaires
	Variant génétique, activité GRK4 γ augmentée	Récepteur D1 dopaminergique	Inhibition de l'activité GRK4 γ rénale
Asthme allergique [37]	Importance de l'arrestine 3 chez l'animal	Récepteurs du chimiotactisme	Inhibition de l'expression de l'arrestine 3
Sclérose en plaques [35]	Baisse de l'expression de la GRK2	Récepteurs de la réponse immune	Inhibition de la GRK2
Maladie de Parkinson [19]	Augmentation de la GRK6 et de l'arrestine 2	Récepteurs dopaminergiques	Inhibition de la GRK6 neuronale
Dépendance aux opiacés [25]	Variation des GRK et des β -arrestines	Récepteurs des opiacés	Inhibition de l'arrestine 3 pour diminuer la tolérance
Cancers hormono-dépendants [30, 31]	Variation de l'expression des GRK	Récepteurs hormonaux	Stimulation de l'expression de la GRK2
Diabète [36]	Importance des GRK chez l'animal	Récepteurs régulant la glycémie	Inhibition de l'activité des GRK2 et -3

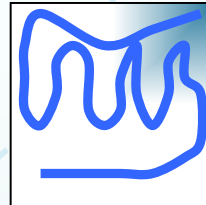
GPCR: non canonical pathway of signal transduction

- *Via* heterotrimeric G proteins
- *independently* from heterotrimeric G proteins



d'après (Orlowski et Grinstein, 1997 -
Lamprecht et al., 1998 - Bockaert et Pin,
1999)

Arrestin
(i3-P)



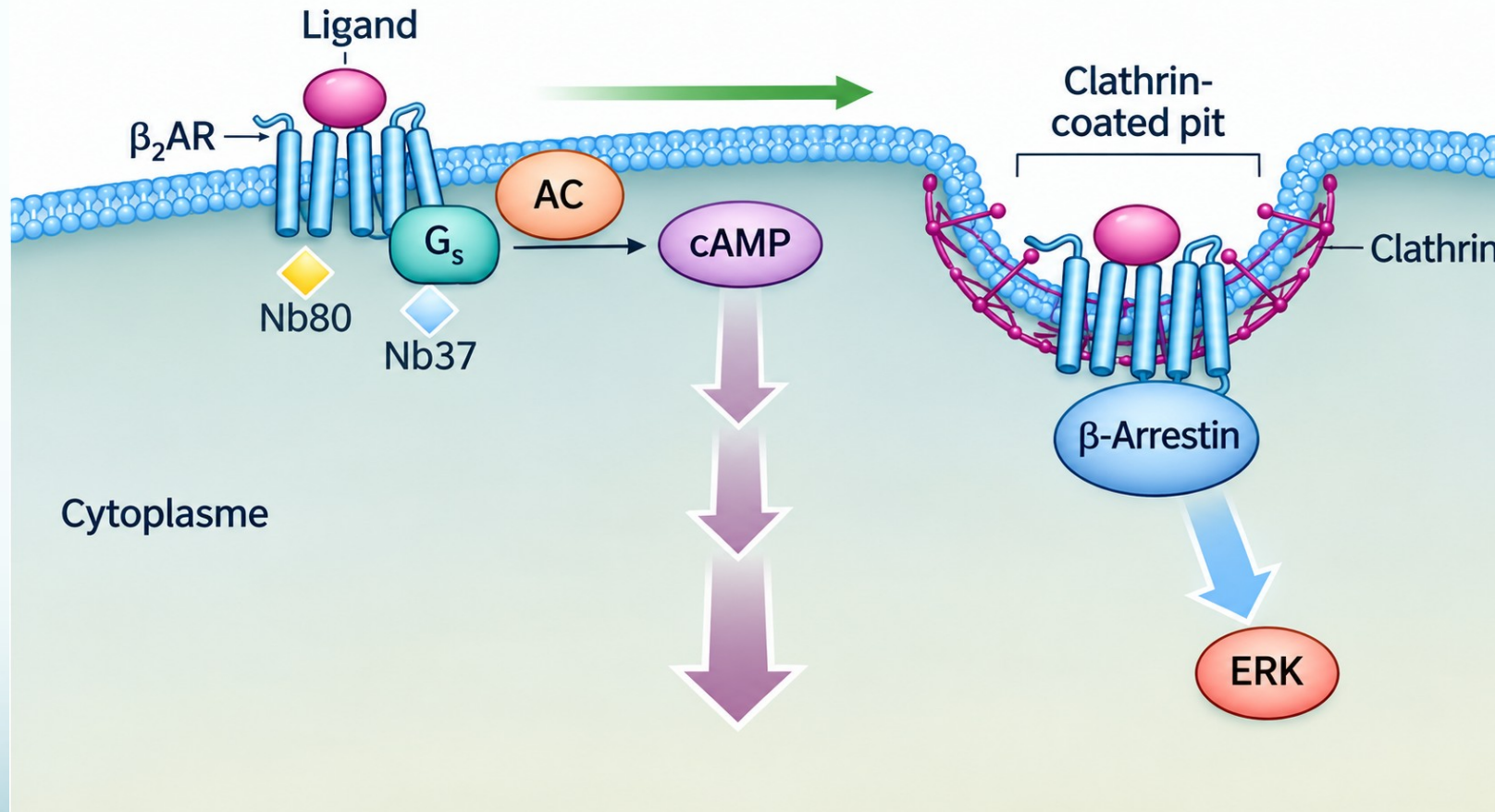
scaffolding proteins
PDZ ex NHERF
(iCT)

scaffolding proteins
~~PDZ ex Homer~~
(iCT)

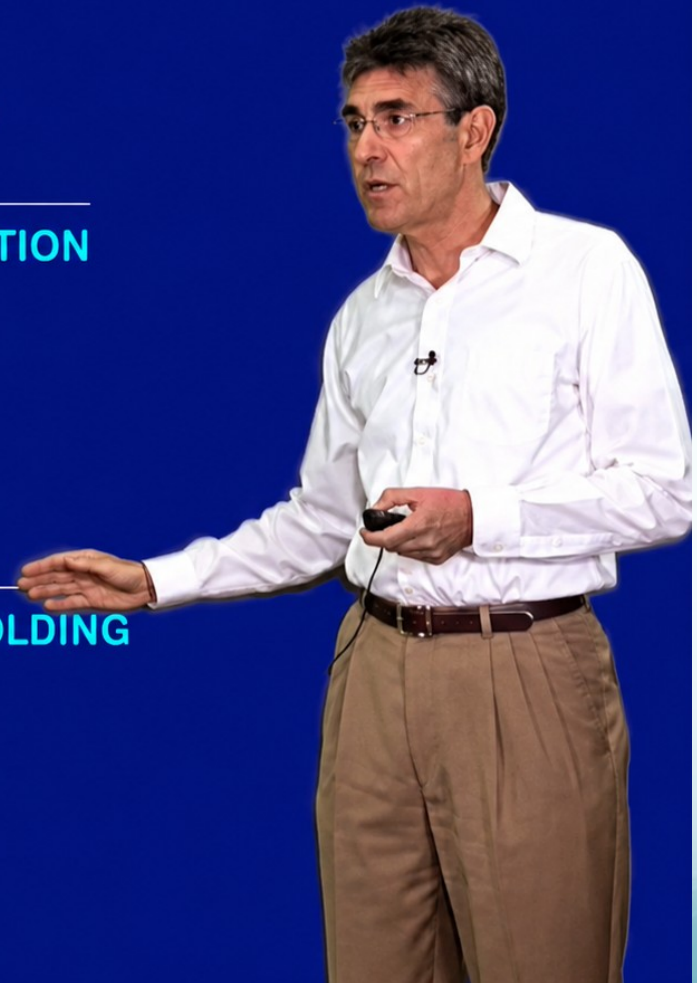
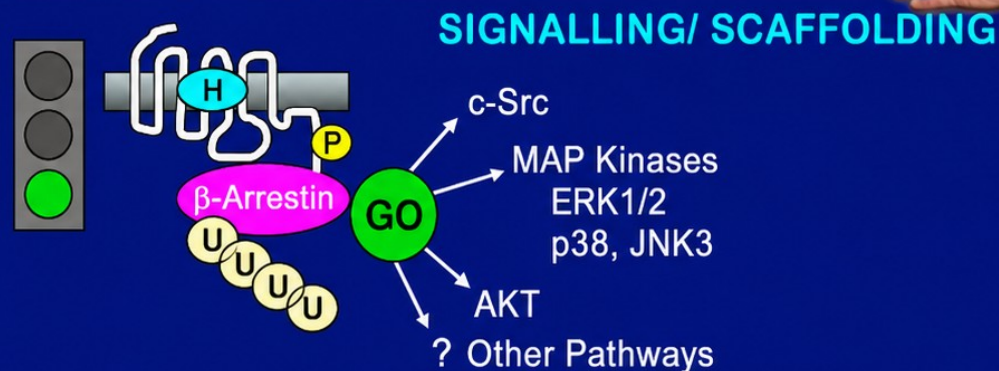
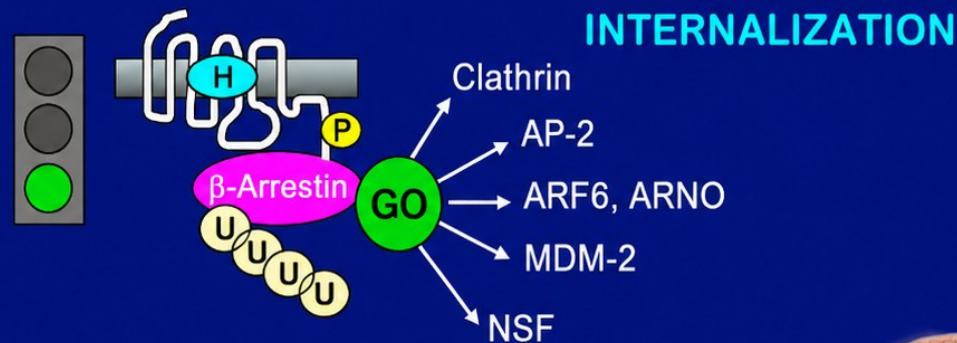
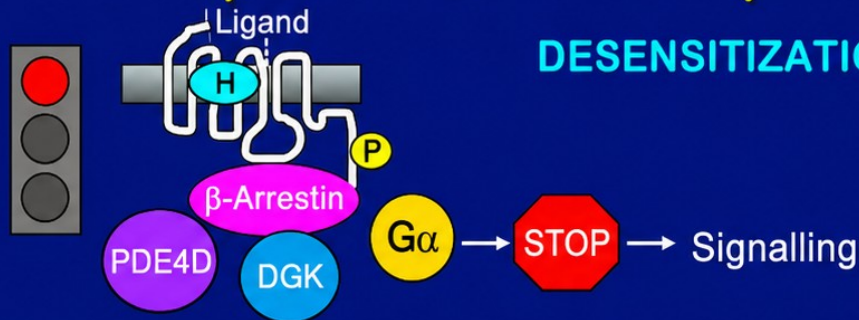
NHERF: Na-H exchanger regulatory factor

GPCR: non canonical pathway of signal transduction

Desensitization and signaling



β -arrestins: Receptor Traffic Cops



Rejoignez le cours de Robert Lefkowitz (Duke University) — Part 2: Beta-arrestin

▶ **EXTRAIT 1** 9:10 → 10:30

▶ **EXTRAIT 2** 12:36 → 22:04

▶ **EXTRAIT 3** 34:05 → 37:55

Sous-titres français demandés à l'ouverture de YouTube (si disponibles).

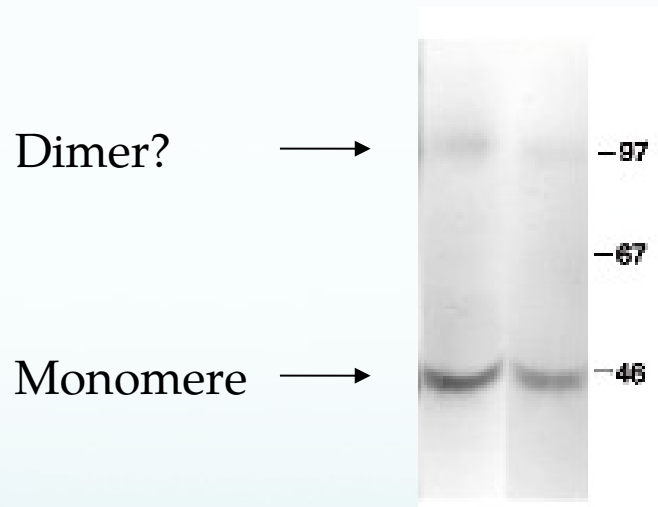


GPCR: exist in the form of dimers

- According to the classical model, GPCRs function as monomers and activate G proteins with a stoichiometry 1Receptor / 1G protein.
- For more than 30 years now, there are experimental evidences that demonstrate that they interact with
 - other proteins than G proteins
 - other GPCR (dimerization or oligomerization)

GPCR: exist in the form of dimers

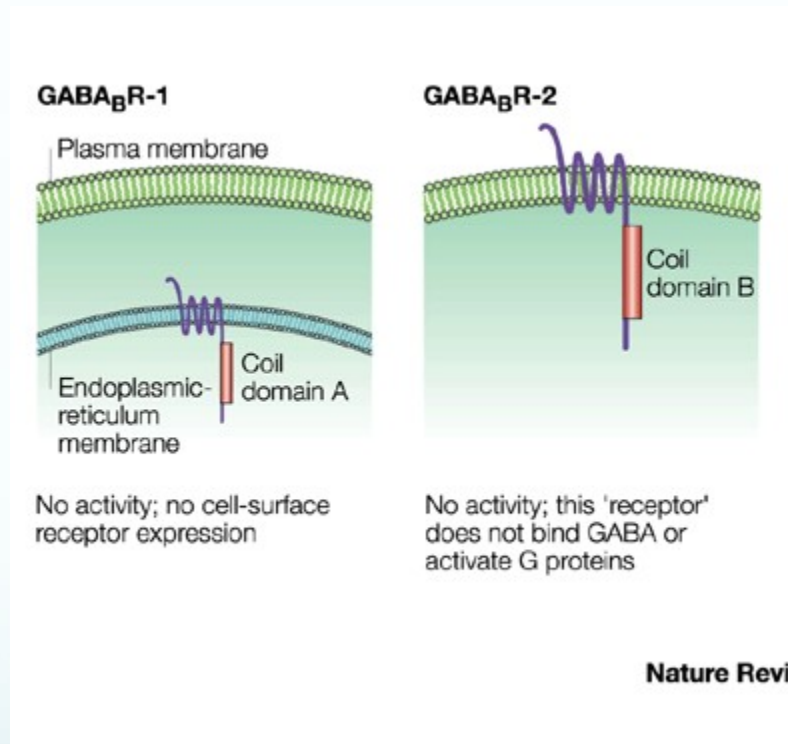
The idea that GPCRs exist in the form of dimers came from the western blot visualizing the expression of the beta-2 adrenergic receptor after transfection of the gene encoding this protein in SF9 cells



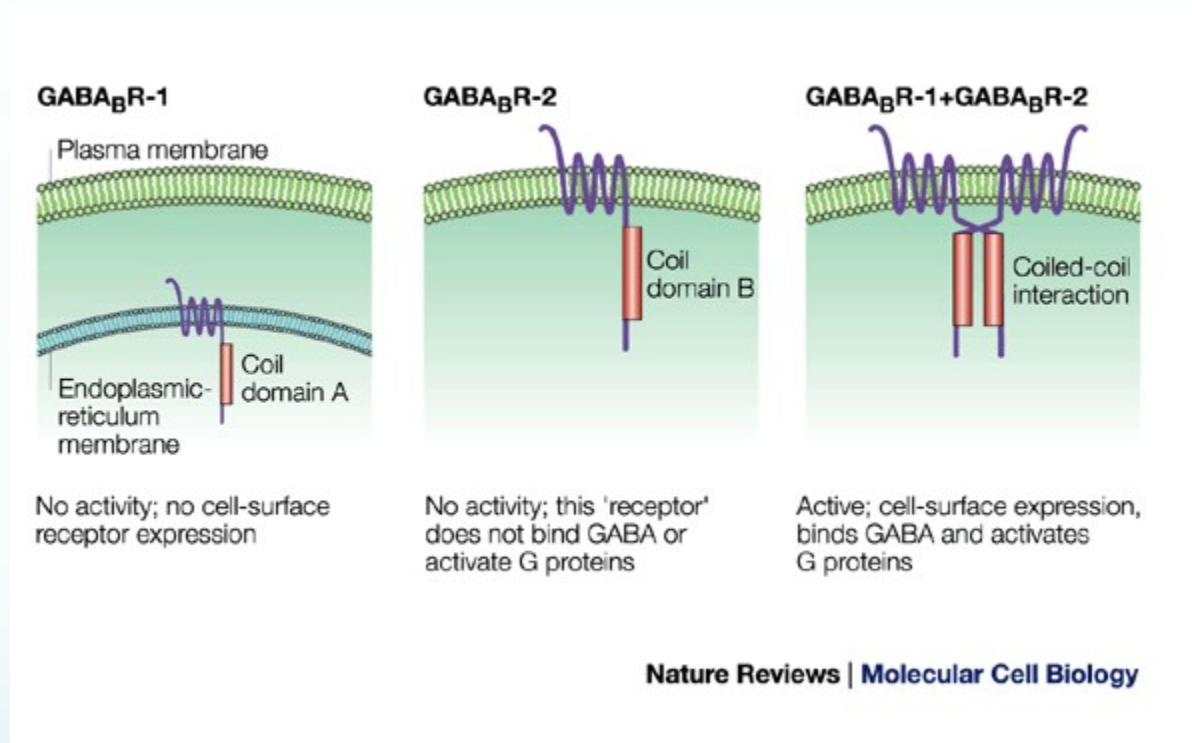
Hebert TE1, Moffett S, Morello JP, Loisel TP, Bichet DG, Barret C, Bouvier M. A peptide derived from a beta2-adrenergic receptor transmembrane domain inhibits both receptor dimerization and activationJ Biol Chem. 1996 271:16384-92.

GPCR: exist in the form of dimers

Also....



GPCR: exist in the form of dimers



- White JH, et al. 1998. Heterodimerization is required for the formation of a functional GABA(B) receptor. *Nature*, 396:679.
- Jones KA, Borowsky B, Tamm JA, Craig DA, Durkin MM, et al. 1998. GABA(B) receptors function as a heteromeric assembly of the subunits GABA(B)R1 and GABA(B)R2. *Nature*.396:674.
- Kaupmann K, et al. 1998. GABA(B)-receptor subtypes assemble into functional heteromeric complexes. *Nature*. 396:683.

GPCR: exist in the form of dimers

Over the last 20 years, this idea has become a working hypothesis and has been demonstrated for many GPCRs using classic techniques to highlight protein-protein interactions such as

- co-immunoprecipitation technique
- FRET or BRET technology,

that you will study in TD.

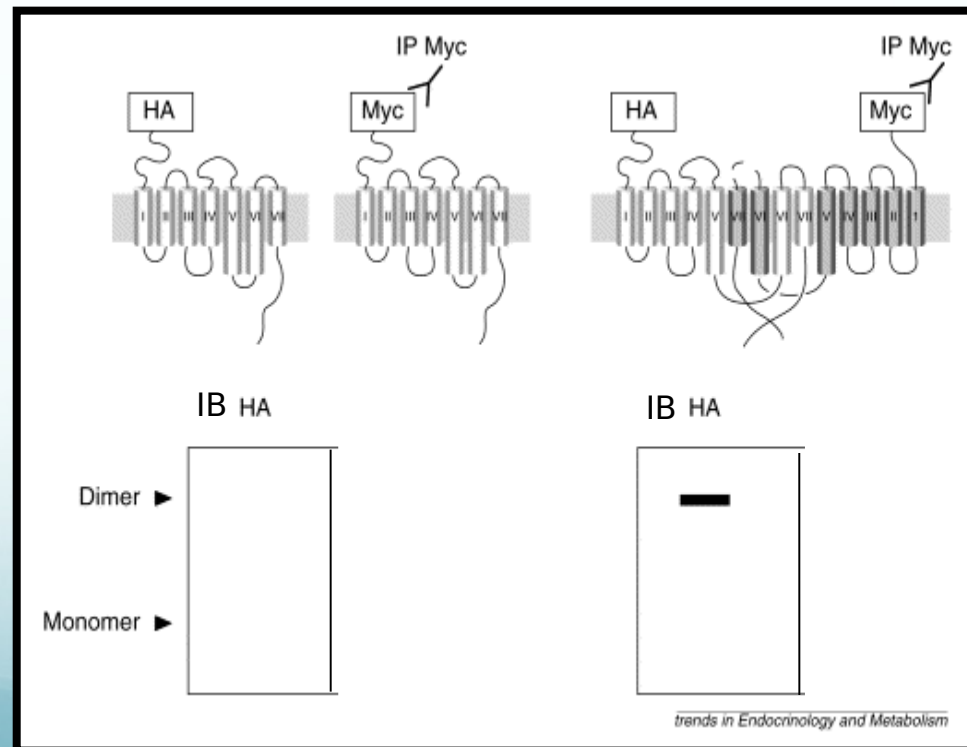
GPCR: exist in the form of dimers

➤ co-immunoprecipitation technology (principle)

IP: immunoprecipitation
with an anti-Myc antibody

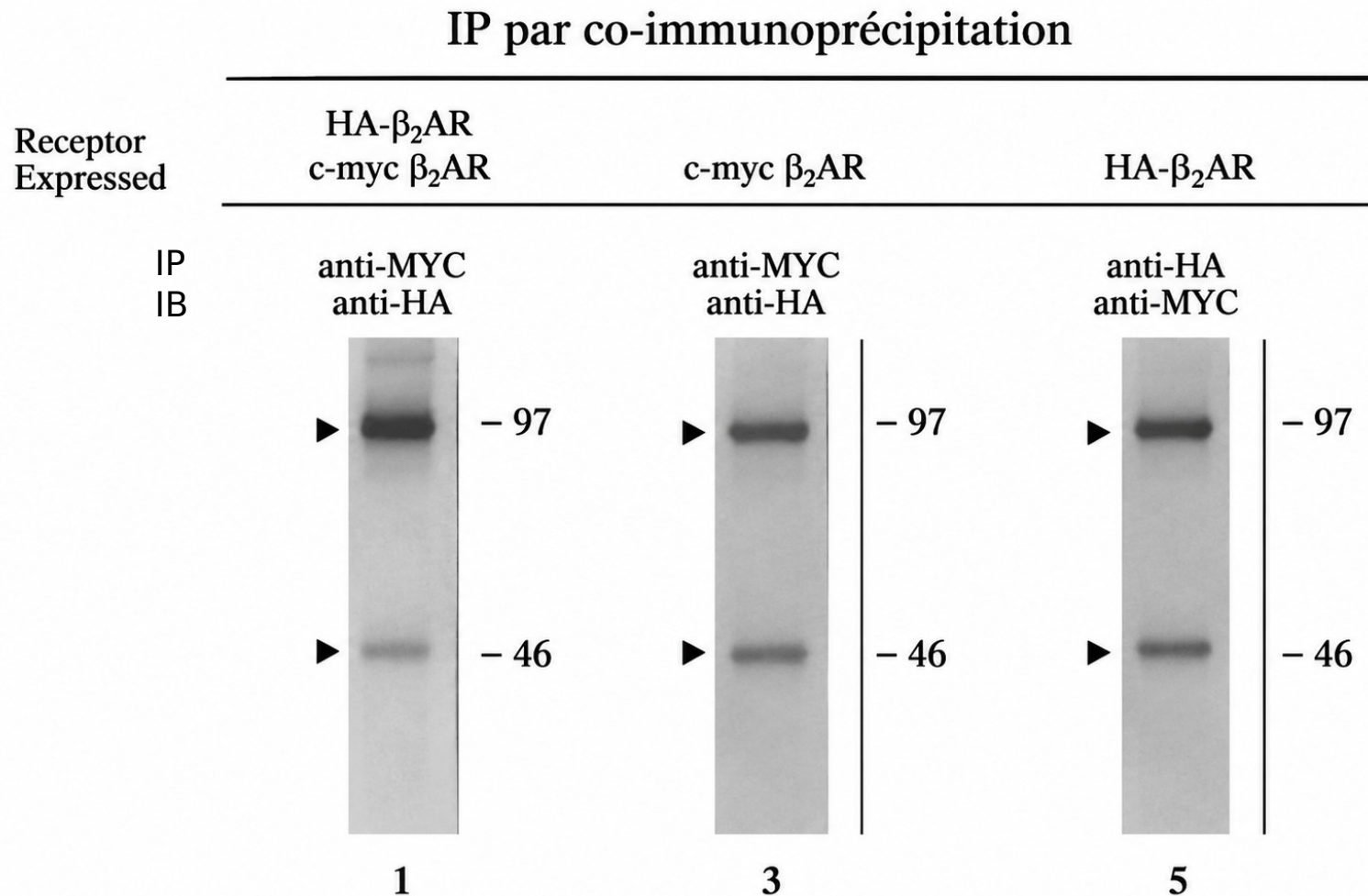
IB: immunoblot with an
anti-HA antibody

HA: haemagglutinin



GPCR: exist in the form of dimers

Demonstration of beta-adrenergic receptors homodimerization by co-immunoprecipitation

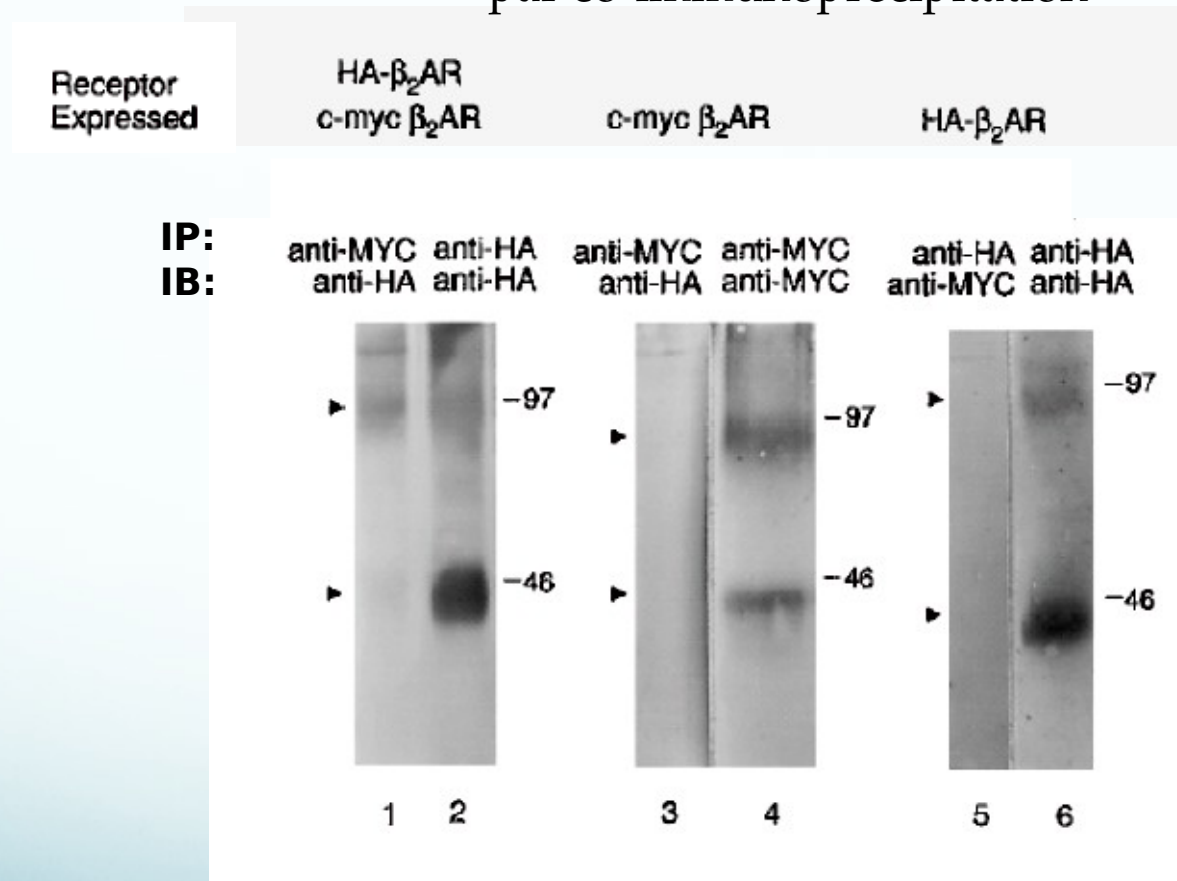


Herbert et al., 1996

Les RCPG: transduction du signal sous la forme de dimères

Les RCPGs existent sous la forme de dimères

Mise en évidence de l'homodimérisation des Récepteurs β_2 -adrénergiques par co-immunoprécipitation



Herbert et al., J Biol Chem, 1996

GPCR: exist in the form of dimers

➤ co-immunoprecipitation technique

Advantages : Direct highlighting

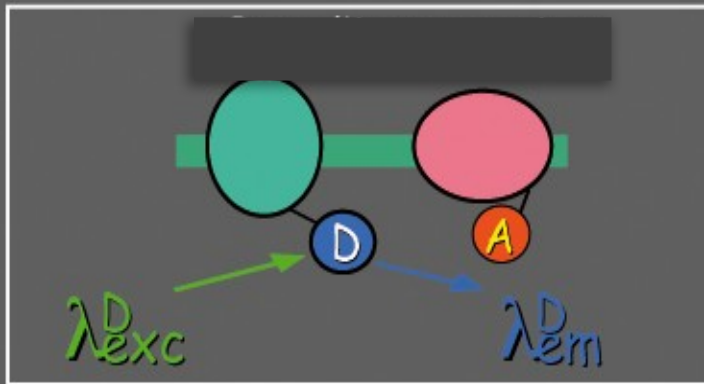
Disadvantages: In vitro • No dynamic study • Aggregation artifact.

GPCR: exist in the form of dimers

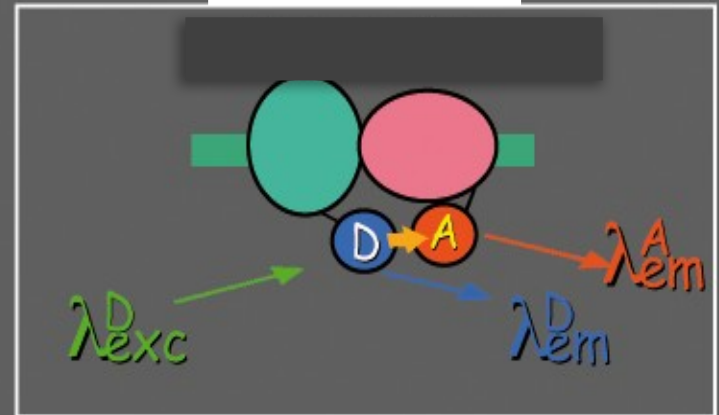
It is the studies carried out in Resonance Energy Transfert that made us accept the idea that GPCRs exist **in living cells** in the form of dimers, homo or heterodimers.

RET: transfert d'énergie par résonance

No interaction



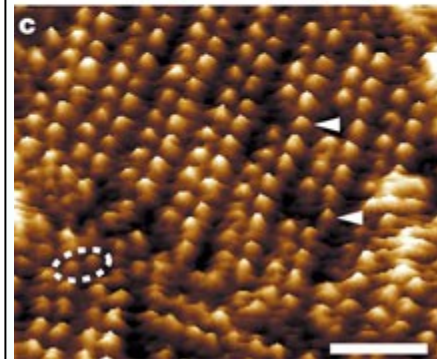
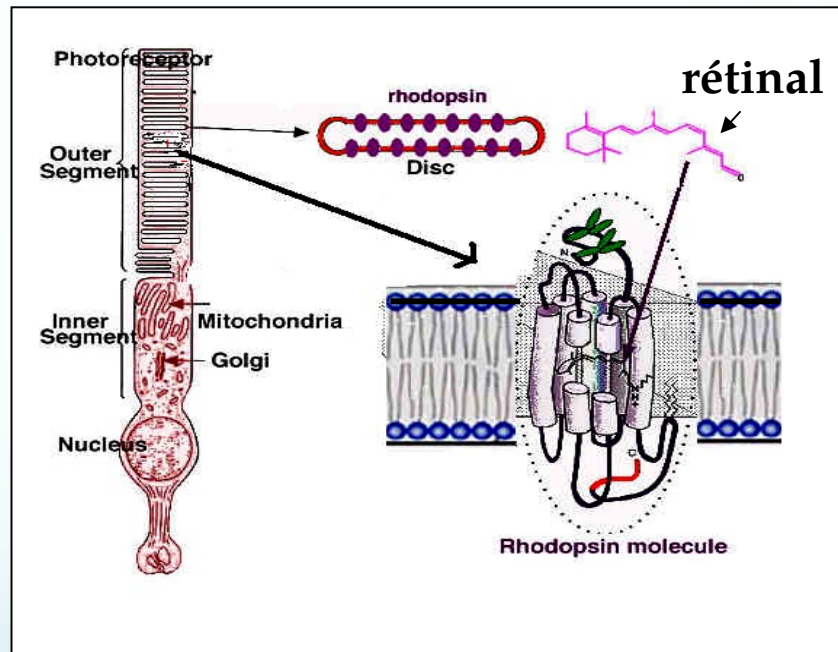
Interaction



GPCR: exist in the form of dimers

Photoreceptor in disks of retinal cell:

*Rhodopsin dimers in
native mouse disc
membranes
(Fotiadis et al. 2003)*



Images obtained by
atomic microscopy

Rhodopsin = protein (**opsin**) + chromophore (**retinal**)

GPCR: exist in the form of dimers

As a result, the recurring question is no longer
"Do the GPCR dimers really exist? "
It has transformed into
"Why GPCRs exist in the form of dimers? "

GPCR: exist in the form of dimers

- Two GPCRs are required for a correct interaction with heterotrimeric G protein. Indeed, the structural data are incompatible with the interaction of a single GPCR with $G\alpha$ and $G\beta\gamma$.
 - The activation of the two constituent protomers of the dimer is necessary for an optimal coupling with the G protein.
 - GPCR dimers are probably the functional unit of signaling
-  Dimerization of GPCRs: to transmit the external signal to the cell.

GPCR: exist in the form of dimers

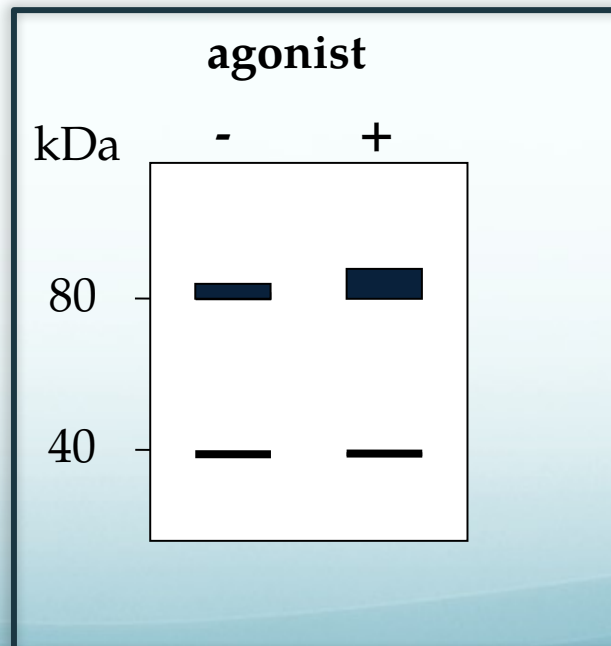
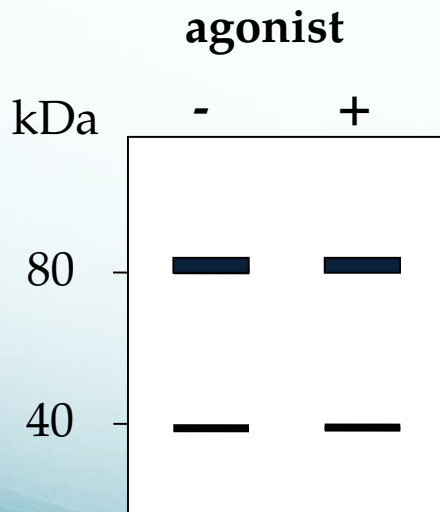
- Is dimerization induced by the agonist?
- * Several articles suggest that dimerization is not constitutive but agonist-induced.
- * On the other hand, others, suggest that GPCRs are in the form of dimers even in the absence of agonists.
- * What is it really?

GPCR: exist in the form of dimers

Is dimerization induced by the agonist?

- Incubation of cells expressing RCPG +/- agonist
- Membrane solubilization
- Immunoblot immunoprecipitated samples analysis

2 possible results

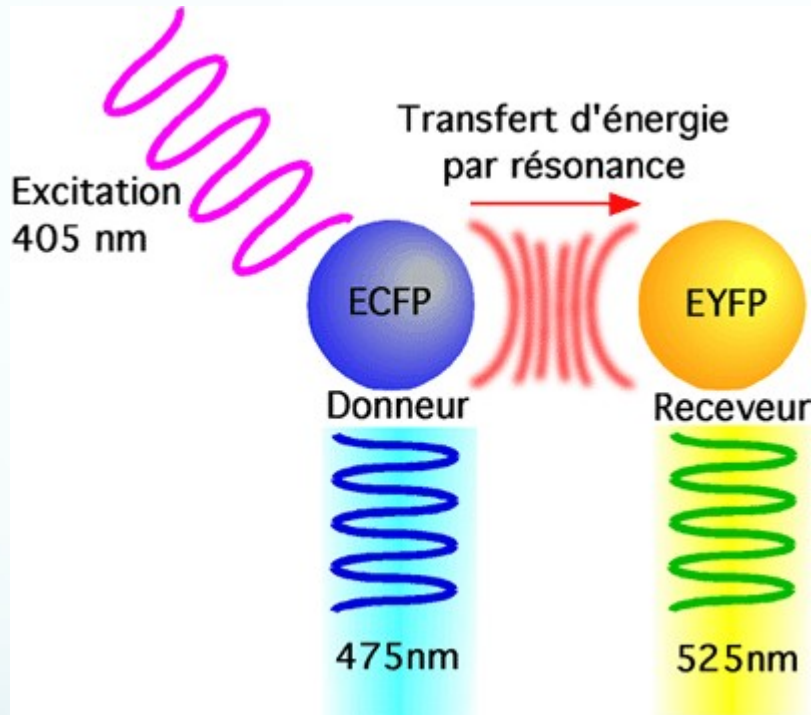


Conclusion 1: the ligand induces receptor dimerization

Conclusion 2: the ligand induces a change of conformation pre-existing dimer as it becomes partially resistant detergent.

GPCR: exist in the form of dimers

Is dimerization induced by the agonist?



If the ligand improves the energy transfer:

Conclusion 1: the ligand induces the formation of dimers.

Conclusion 2: the agonist, by binding, modifies the conformation of the R already in the form of dimer and brings the donor molecule closer and the acceptor molecule.

GPCR: exist in the form of dimers

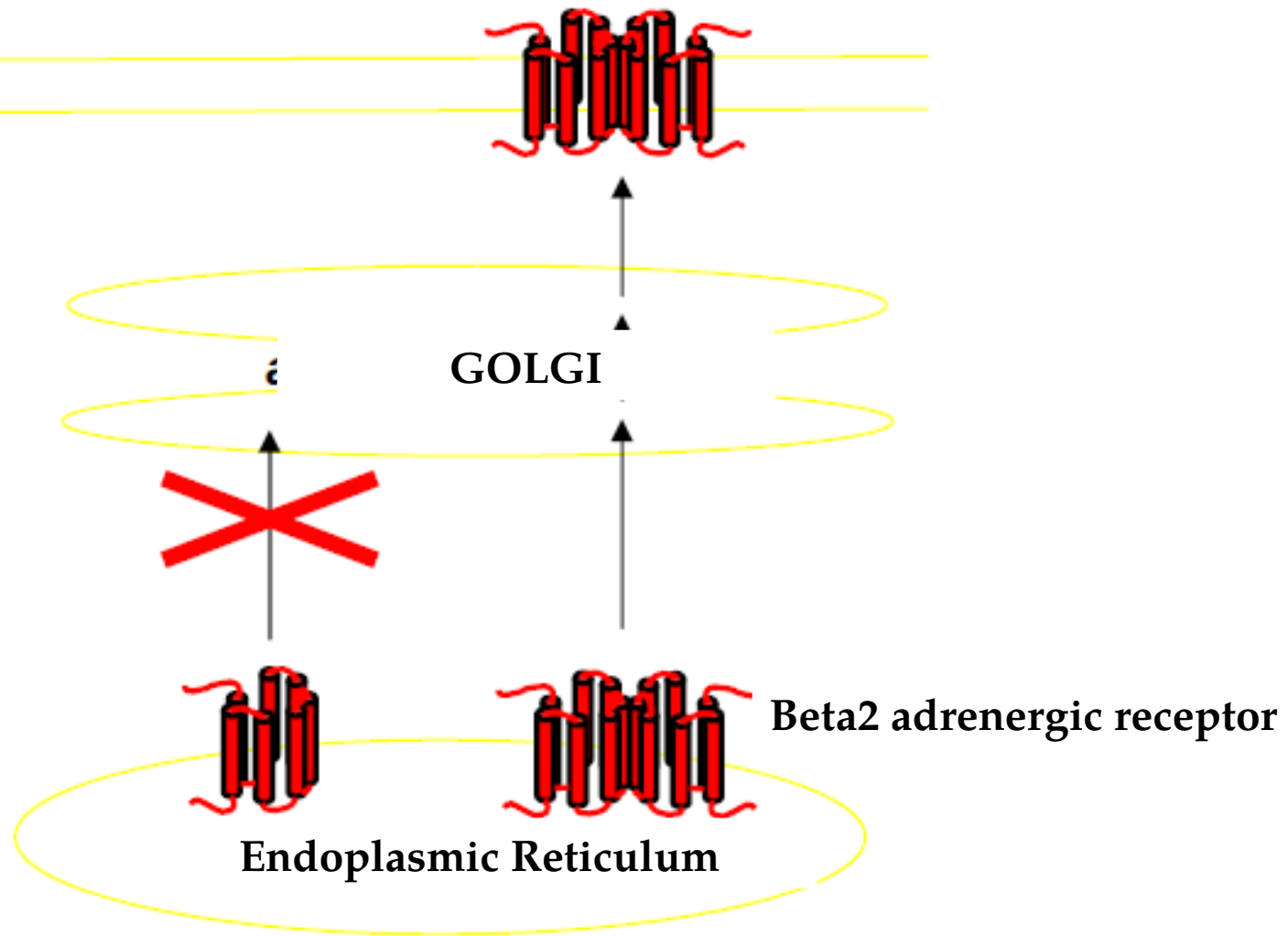
Is dimerization induced by the agonist?

- No
- Because there is no data that unequivocally demonstrates that ligand interaction is required for dimerization of GPCRs
- Most importantly, most studies indicate that dimers form early during receptor synthesis upon entry into the ER.

GPCR: exist in the form of dimers

- The formation of homo- or heterodimers: a prerequisite for their export to the plasma membrane?
- This hypothesis is consistent with the fact that most membrane proteins require oligomerization to be properly processed and sent to the plasma membrane.

Consequence on the routing of GPCRs



Salahpour et al., JBC, 2004

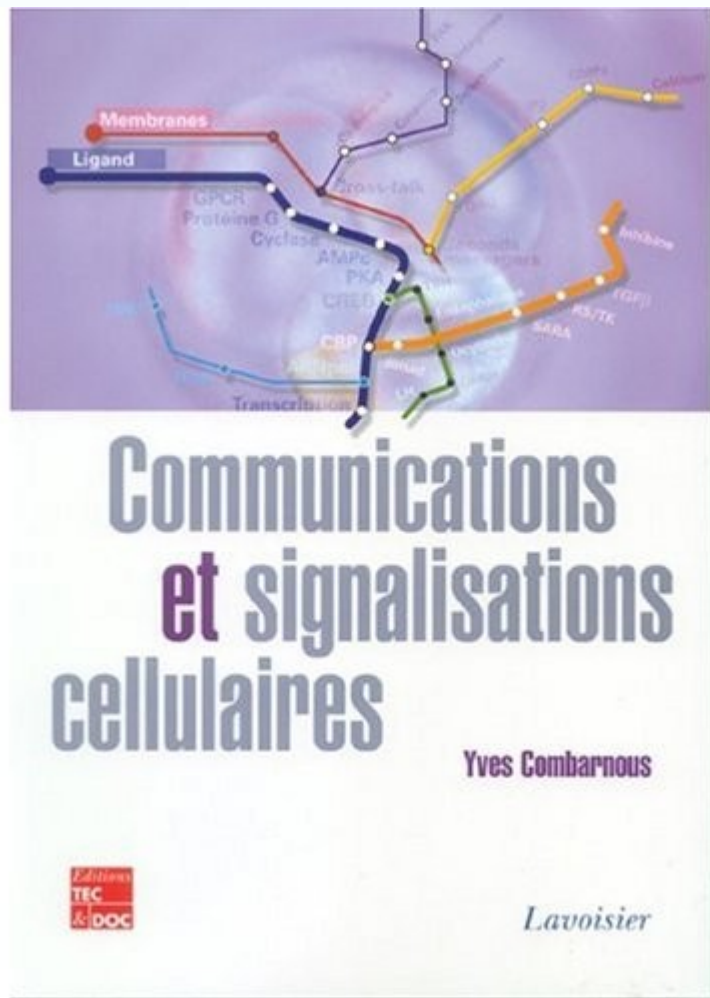
Tableau 1 : Oligomérisation des RCPG

structure	récepteur	références
dimérisation 7 TM/7 TM	β_2 -adrénergique	Hebert et al., 1996 - Hebert et al., 1998
	histaminique H_2	Fukushima et al., 1997a
	dopaminergiques D_1 , D_2 et D_3	Ng et al., 1996 - Nimchinsky et al., 1997 - George et al., 1998
	muscariniques M_2 et M_3	Potter et al., 1991 - Maggio et al., 1993 - Maggio et al., 1996 - Maggio et al 1999 - Zeng et Wess, 1999
	adénosine A_1	Ciruela et al., 1995
	sérotoniques $5HT_{1B}$ et $5HT_{1D}$	Xie et al., 1999
	angiotensine II AT_{1A}	Monnot et al., 1996
	bradykinine B_2	Abdalla et al., 1999
	neurokinines NK-1 et NK-2	Huang et al., 1994 - Huang et al., 1995a
	vasopressine V_2	Hebert et al., 1996 - Zhu et Wess, 1998
	opioïdes δ et κ	Cvejic et Devi, 1997 - Jordan et Devi, 1999
	acide γ -aminobutyrique type B $GABA_B R1$ et $GABA_B R2$	Jones et al., 1998 - Kaupmann et al., 1998 - White et al., 1998
	métabotropiques de l'acide glutamique $mGluR1$ et $mGluR5$	Romano et al., 1996 - Okamoto et al., 1998a
	"calcium-sensing" (CaR)	Bai et al., 1998 - Bai et al., 1999
	somatostatine ssr_5 et ssr_1	Rocheville et al., 2000
dimérisation 7 TM/1 TM	visuels (rhodopsine des bâtonnets et opsines des cônes)	Baker et al., 1994 - Ferreira et al., 1996 Ferreira et al., 1997
	olfactifs	Dwyer et al., 1998
	calcitonine (CTR)	Christopoulos et al., 1999
	"calcitonin-receptor-like-receptor" (CRLR)	McLatchie et al., 1998 - Buhlmann et al., 1999 - Kamitani et al., 1999
dimérisation 7 TM/récepteurs- canaux	dopamine D_5	Liu et al., 2000

GPCR: exist in the form of dimers

Conclusions

- GPCRs exist as homo- and hetero-dimers
 - Dimerization takes place in the ER
- Once in the membrane, the dimer is the functional unit that recruits G protein
- At the molecular level, it remains to define the structure of heterodimers and the rules that allow some protomers to associate.
- At the functional level, it remains to define the physiological consequences of homodimerization and heterodimerization and to see to what extent we can use this process to develop new therapeutic molecules



PHARMACOLOGIE

Des cibles vers l'indication
thérapeutique

Yves Landry
Jean-Pierre Gies

DUNOD